

Microprocessor Systems

Lab0 – Environment Setup

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NTHU EE

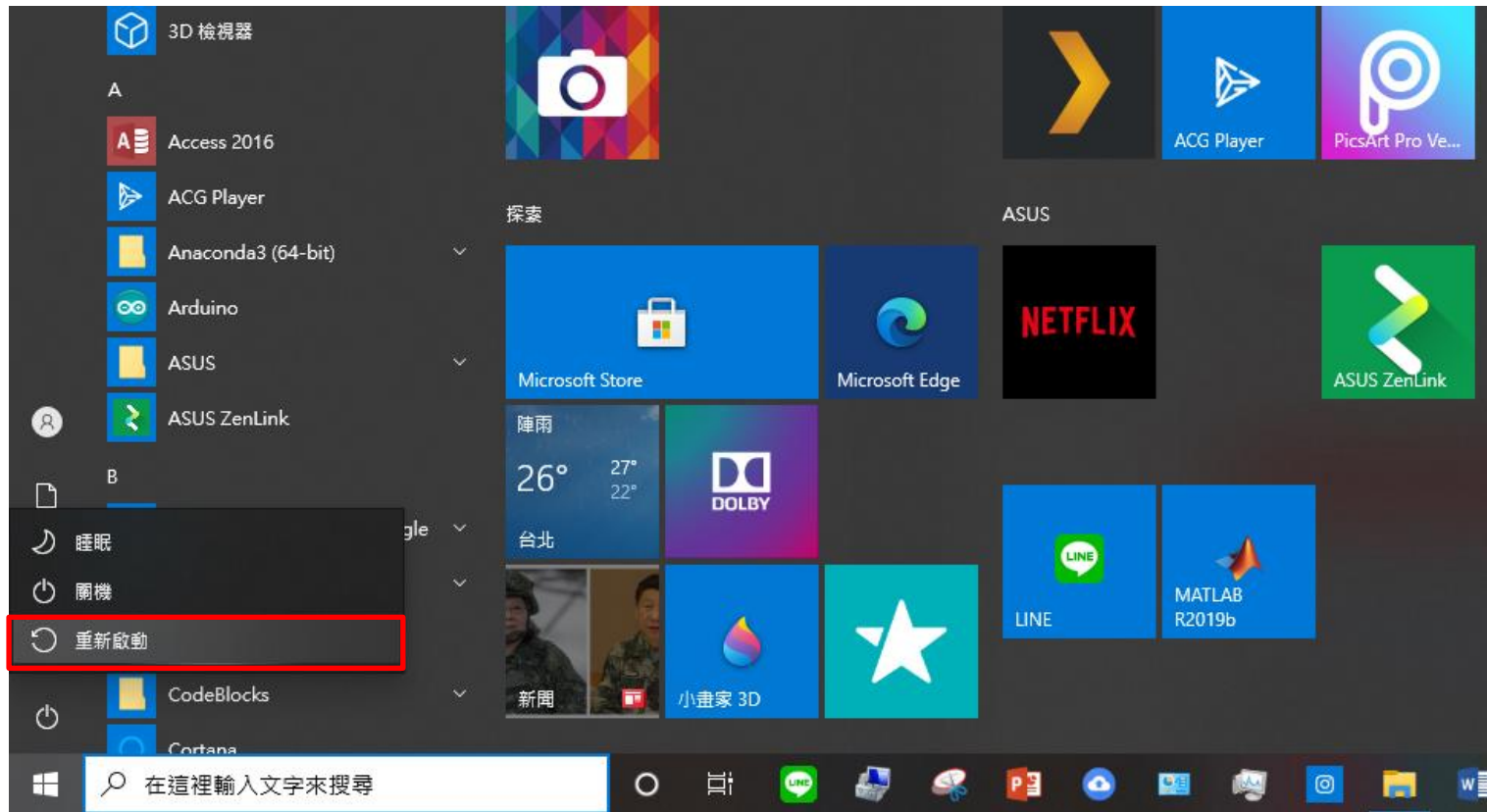


Outline

- Environment Setup
 - USB driver installation(Win10)
 - USB driver installation (Win7)
 - Visual Studio Code installation
 - SDCC installation
 - MinGW installation
 - ASEM-51 installation
 - Add to Path
 - Add “make” command
 - How to make project
 - How to program .hex into 8051

USB driver installation Win10

USB driver installation (Win10)



Press “Shift” and restart the computer

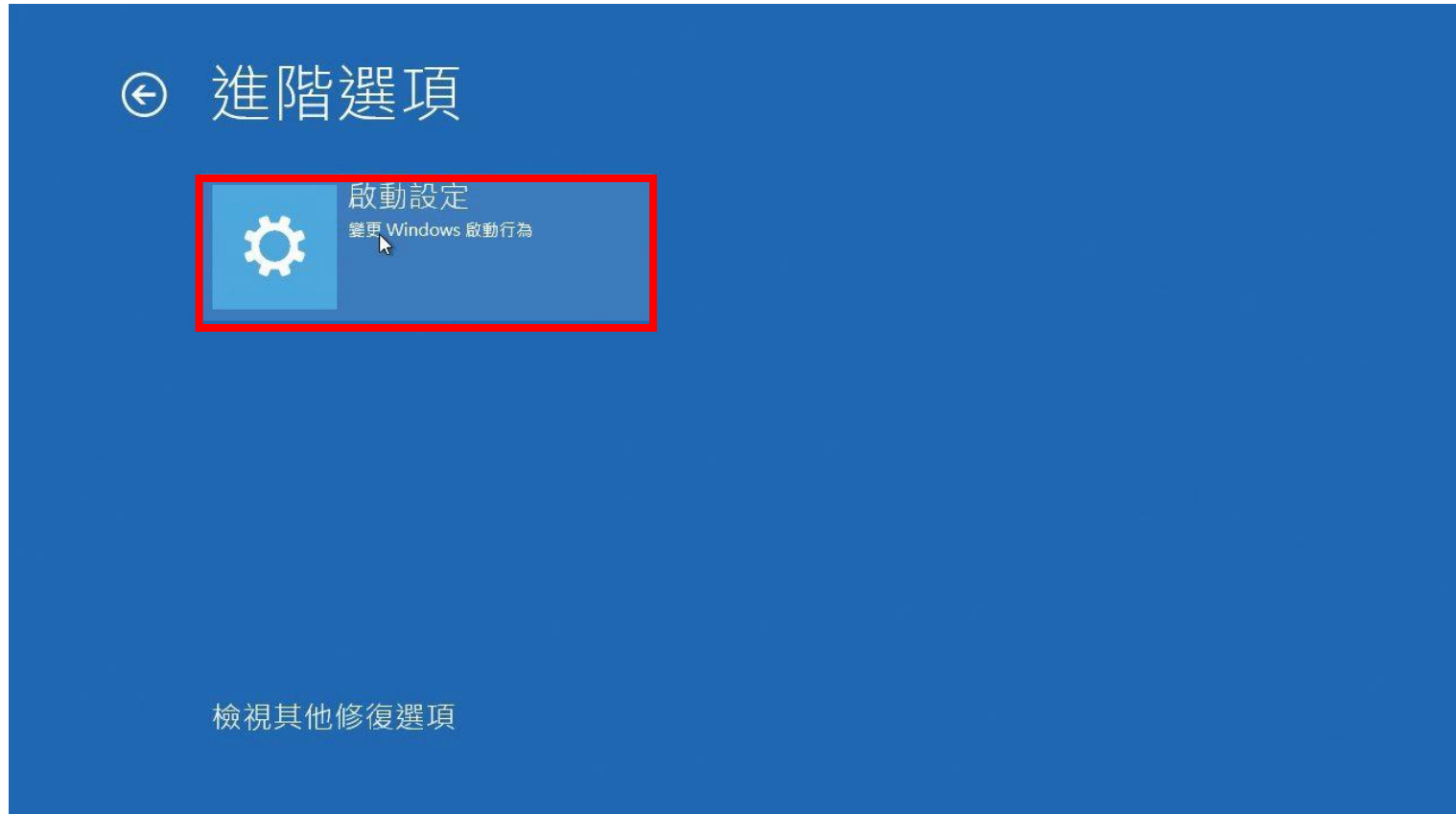
USB driver installation (Win10)



USB driver installation (Win10)



USB driver installation (Win10)



USB driver installation (Win10)



USB driver installation (Win10)

啟動設定

按下數字以選擇下面的選項:

使用數字鍵或功能鍵 F1-F9 .

- 1) 啟用偵錯
- 2) 啟用開機記錄
- 3) 啟用低解析度視訊
- 4) 啟用安全模式
- 5) 啟用安全模式 (含網路功能)
- 6) 啟用安全模式 (含命令提示字元)
- 7) 停用驅動程式強制簽章
- 8) 停用開機初期啟動的反惡意程式碼保護
- 9) 停用失敗時自動重新啟動

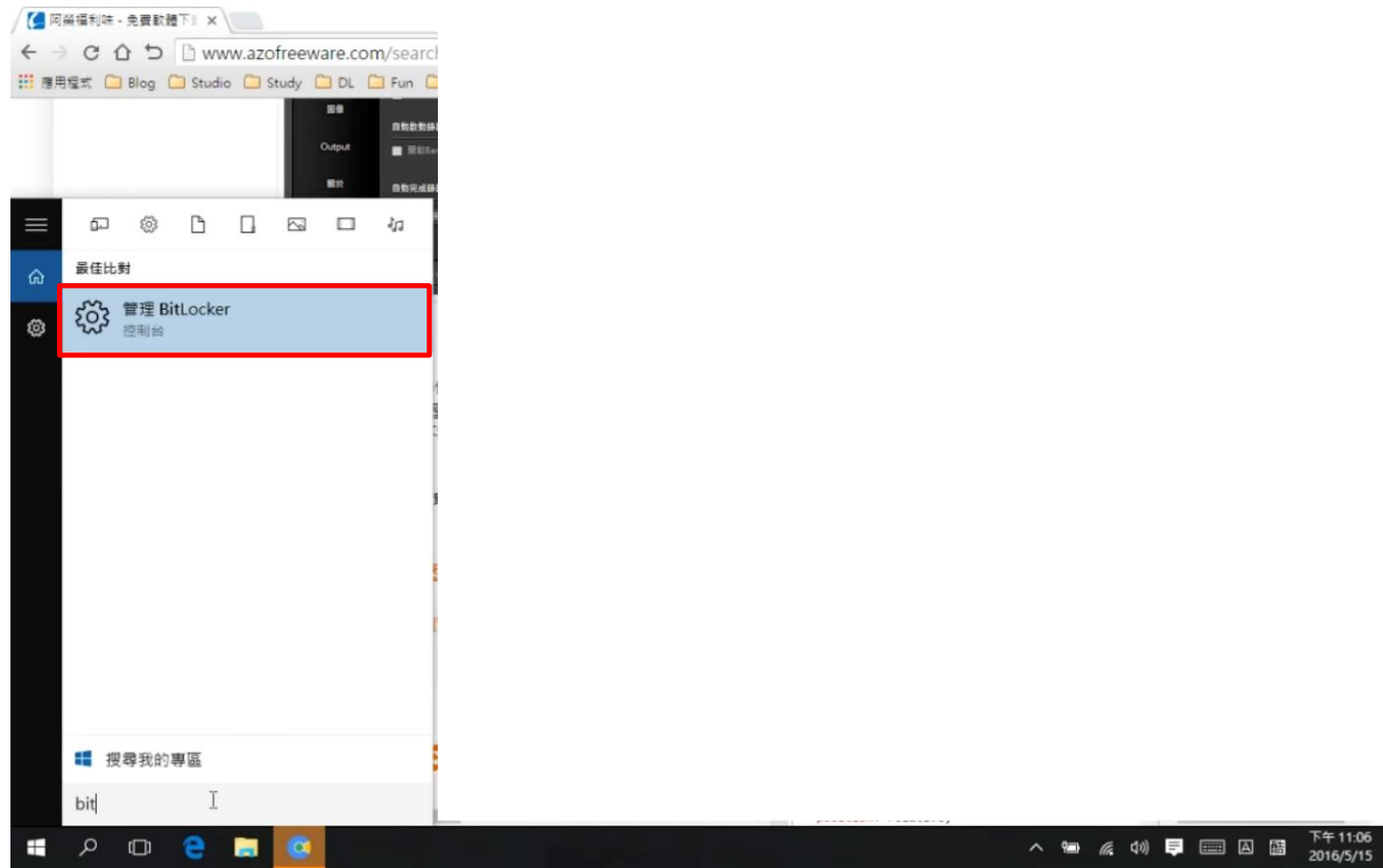
按下 F10 檢視其他選項

按下 Enter 以返回作業系統

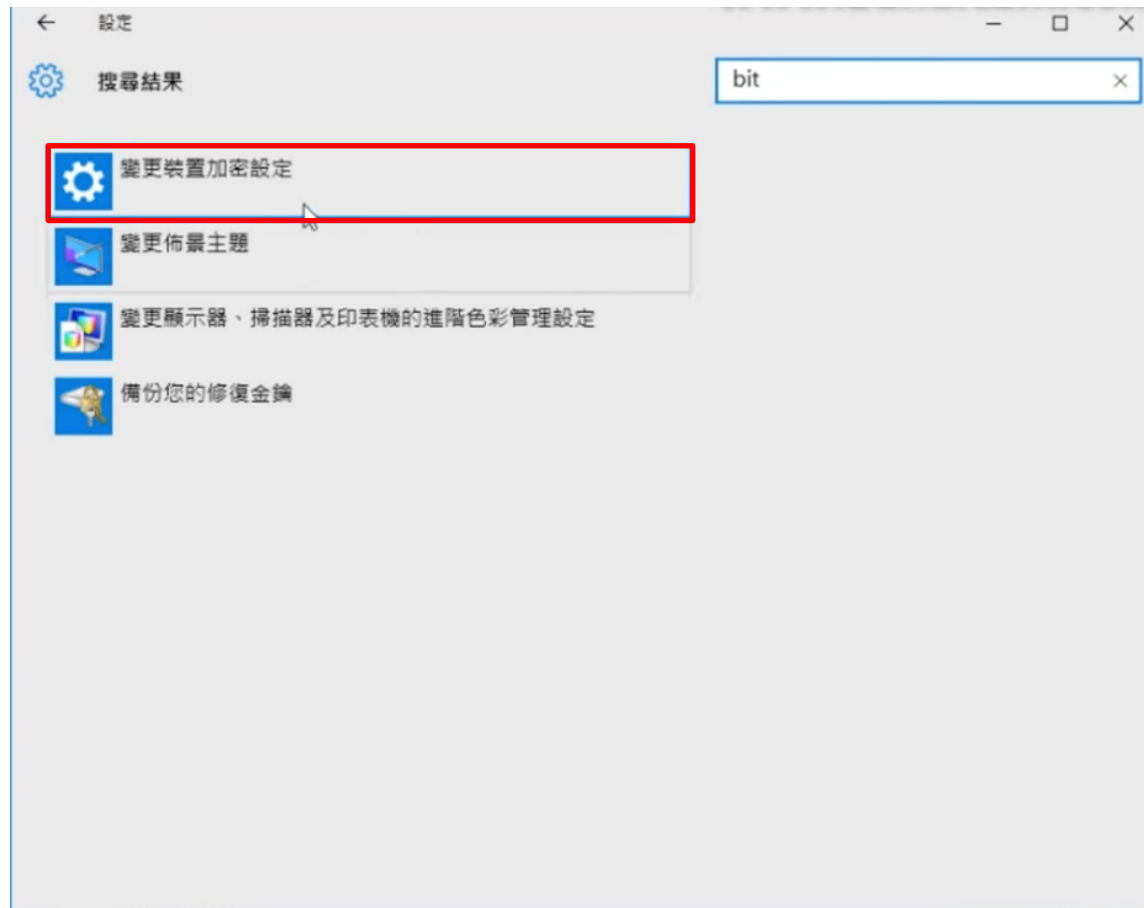
Press“7”-> 停用驅動程式強制簽章

**If you have the issue with “BitLocker”
while “停用驅動程式強制簽章”**

Search “bitlocker”



Choose “變更加密設定” (differ from OS version)



Turn off “裝置加密”



After complete, run p.9 again



USB driver installation (Win10)



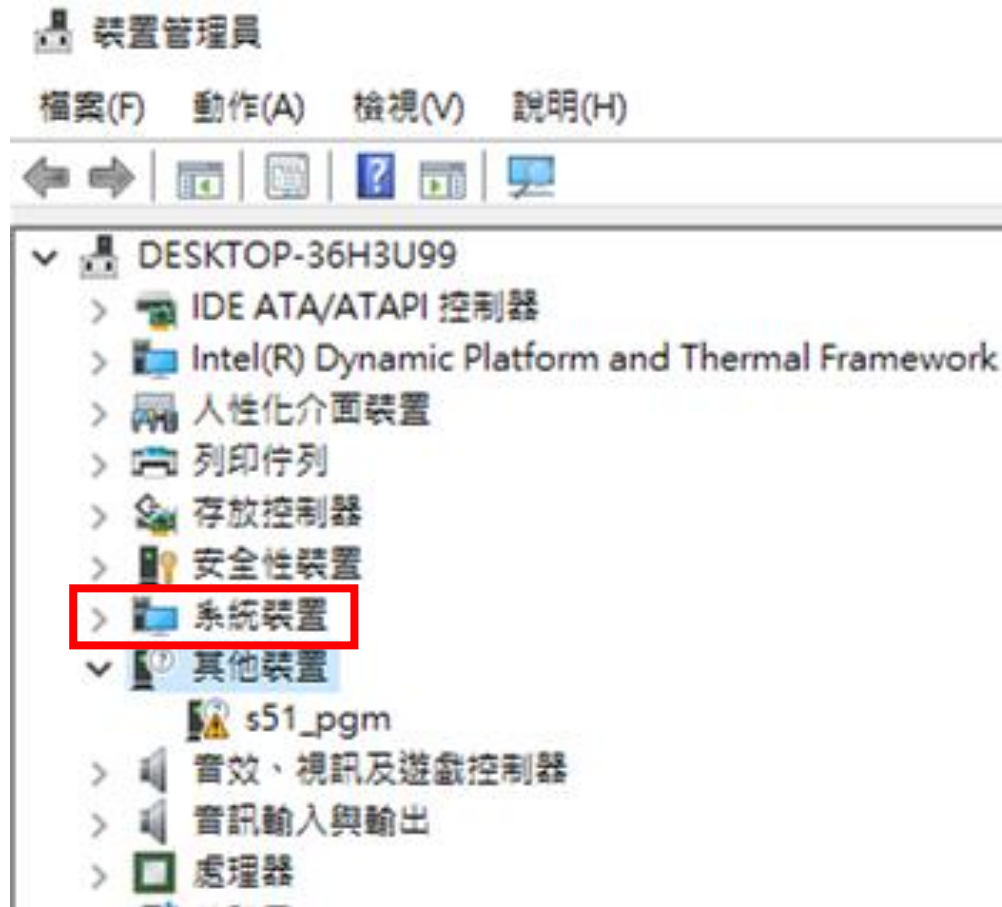
After restarting computer,
connect USB port of KT89S51 to the computer

USB driver installation (Win10)



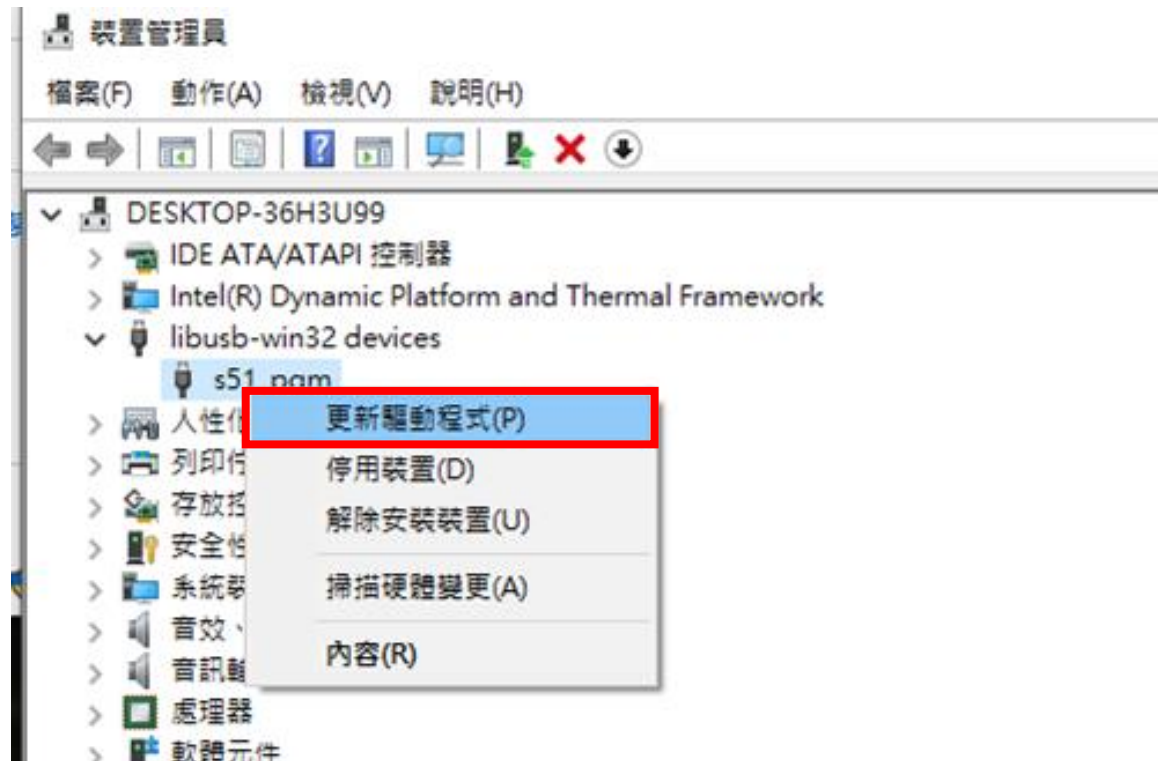
Open 裝置管理員 (device manager)

USB driver installation (Win10)



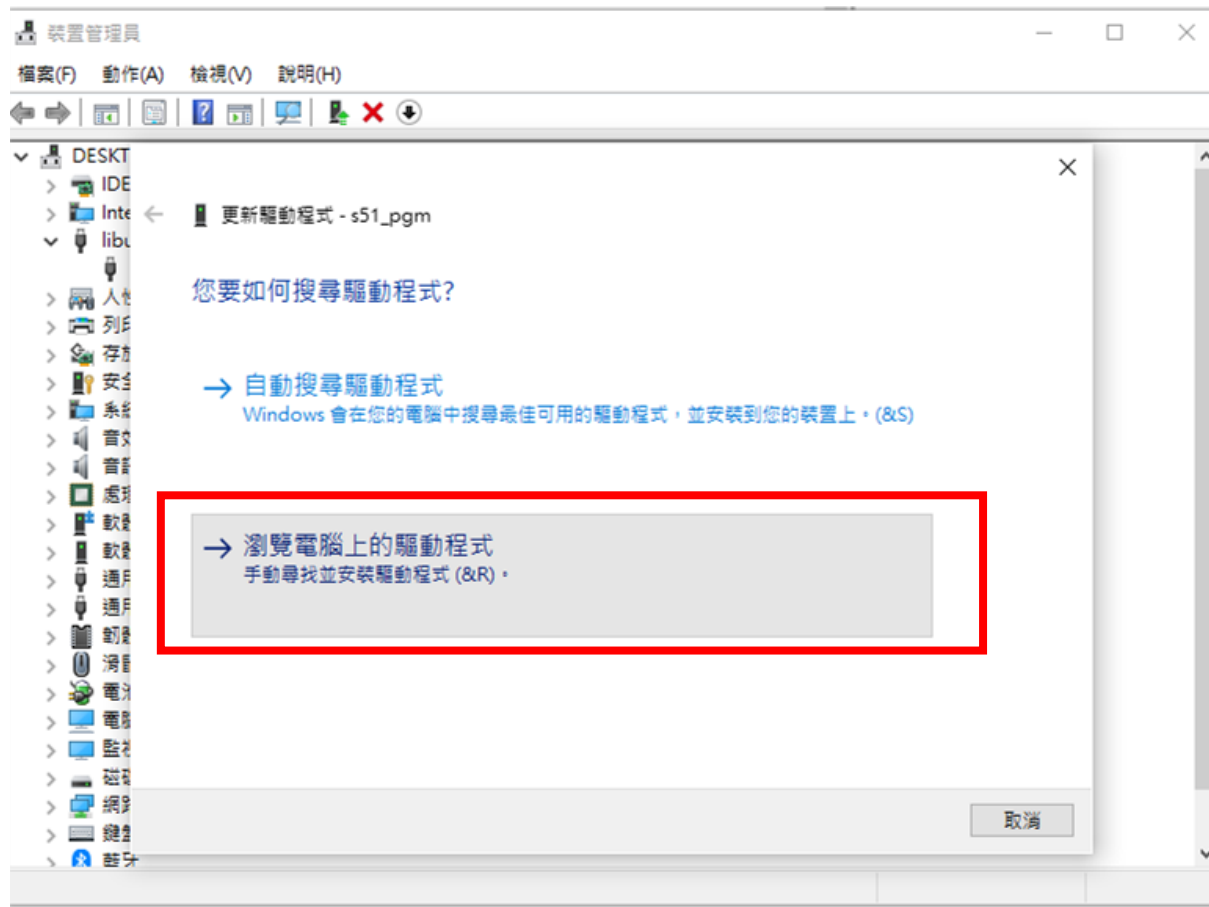
Click “其他裝置”

USB driver installation (Win10)



Click “s51_pgm” and right click,
and then “更新驅動程式”

USB driver installation (Win10)



USB driver installation (Win10)

更新驅動程式 - s51_pgm

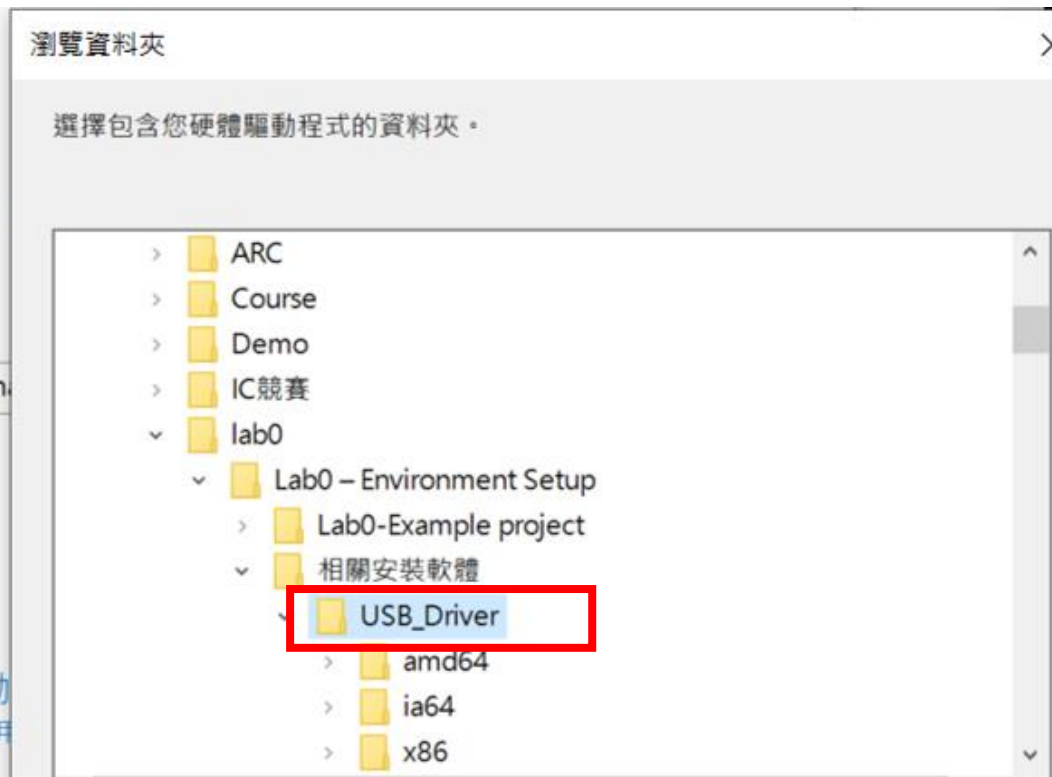
在您的電腦上瀏覽驅動程式

在此位置搜尋驅動程式:

C:\Program Files (x86)\Edimax\Edim

☒ 包含子資料夾(I)

→ 讓我從電腦上的可用驅動
此清單將會顯示與裝置相容的可用

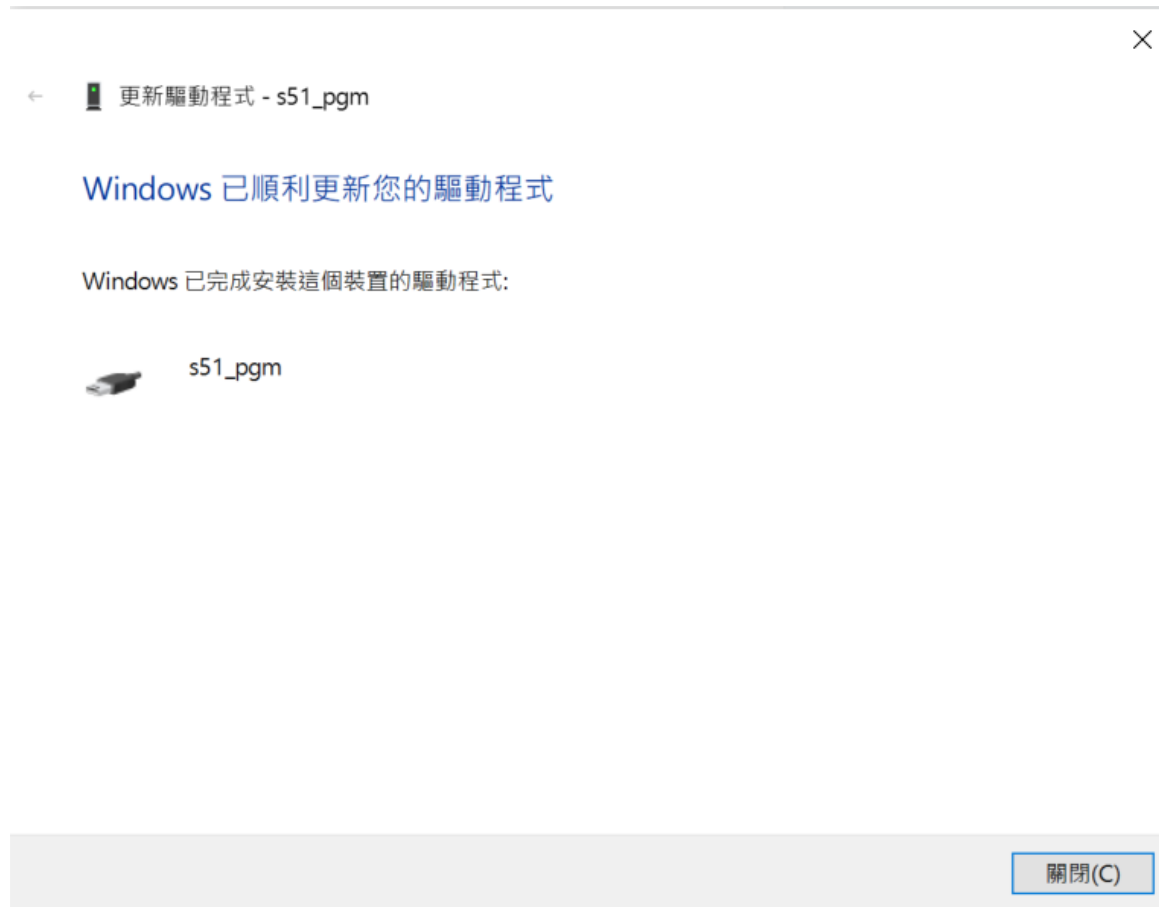


Click "瀏覽", choose the path of USB driver
(Go to google drive and download Related_File.zip)

USB driver installation (Win10)



USB driver installation (Win10)



Click“關閉”, restart the computer.

USB driver installation

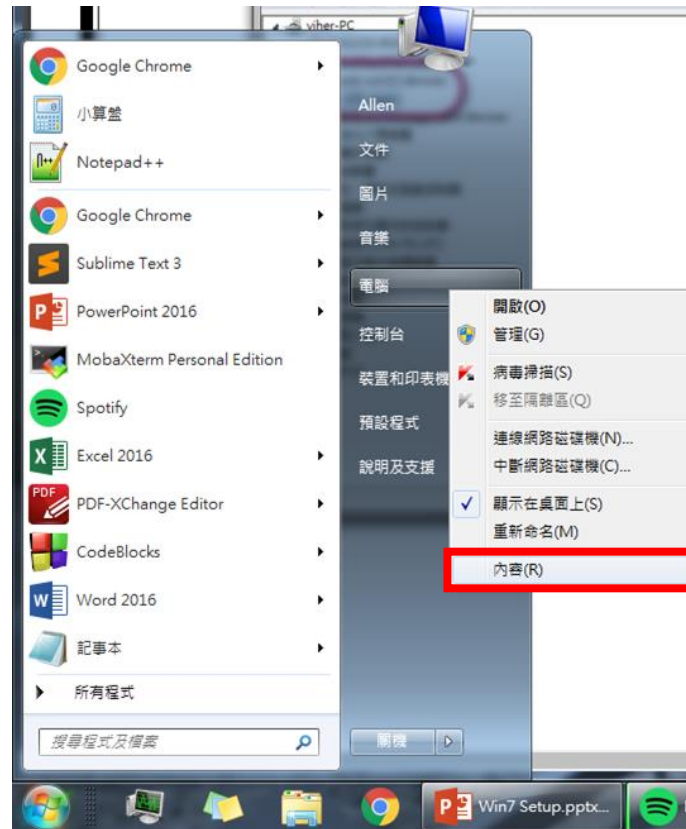
Win7

USB driver installation (Win7)



After restarting computer,
connect USB port of KT89S51 to the computer

USB driver installation (Win7)



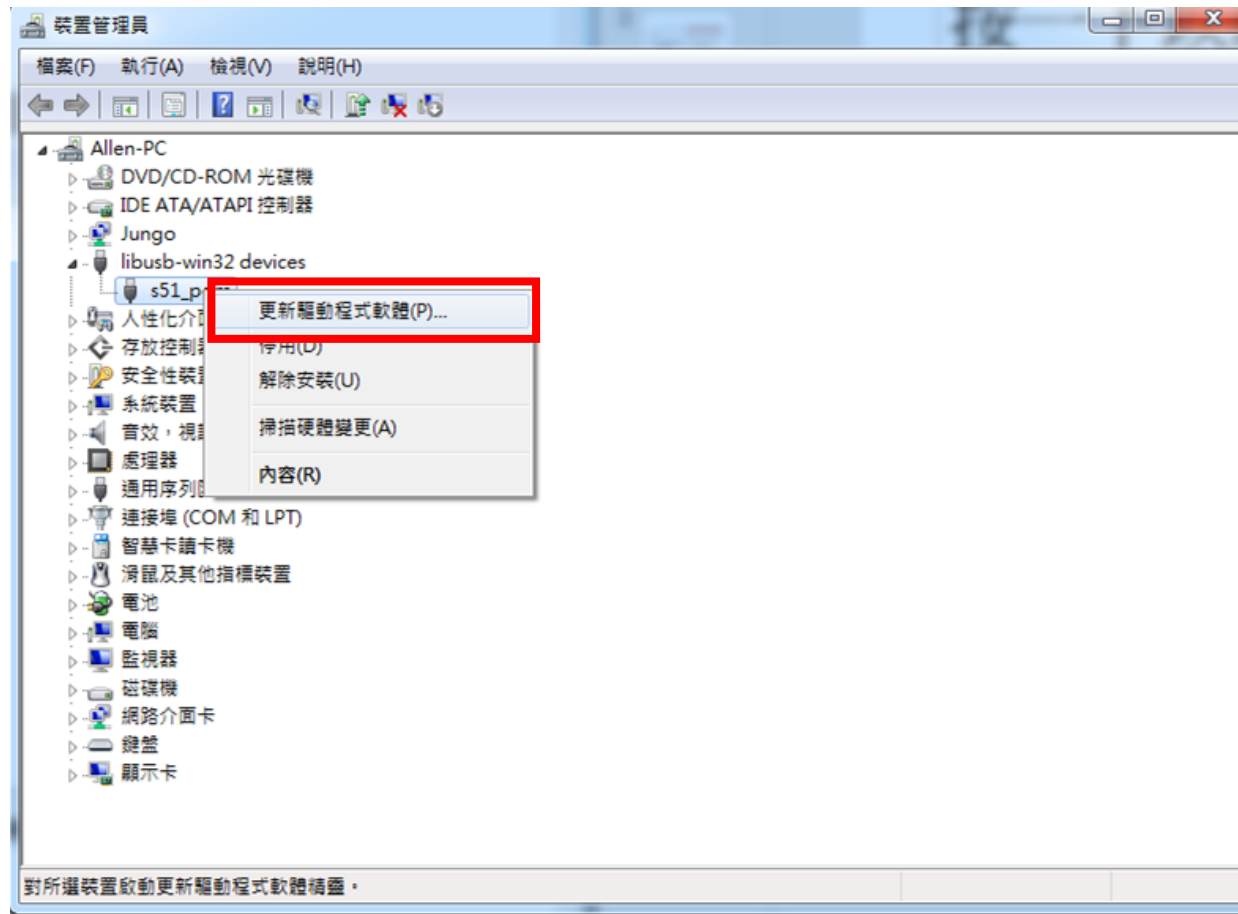
Click “windows”, move to “電腦”, right click and then choose “內容”

USB driver installation (Win7)



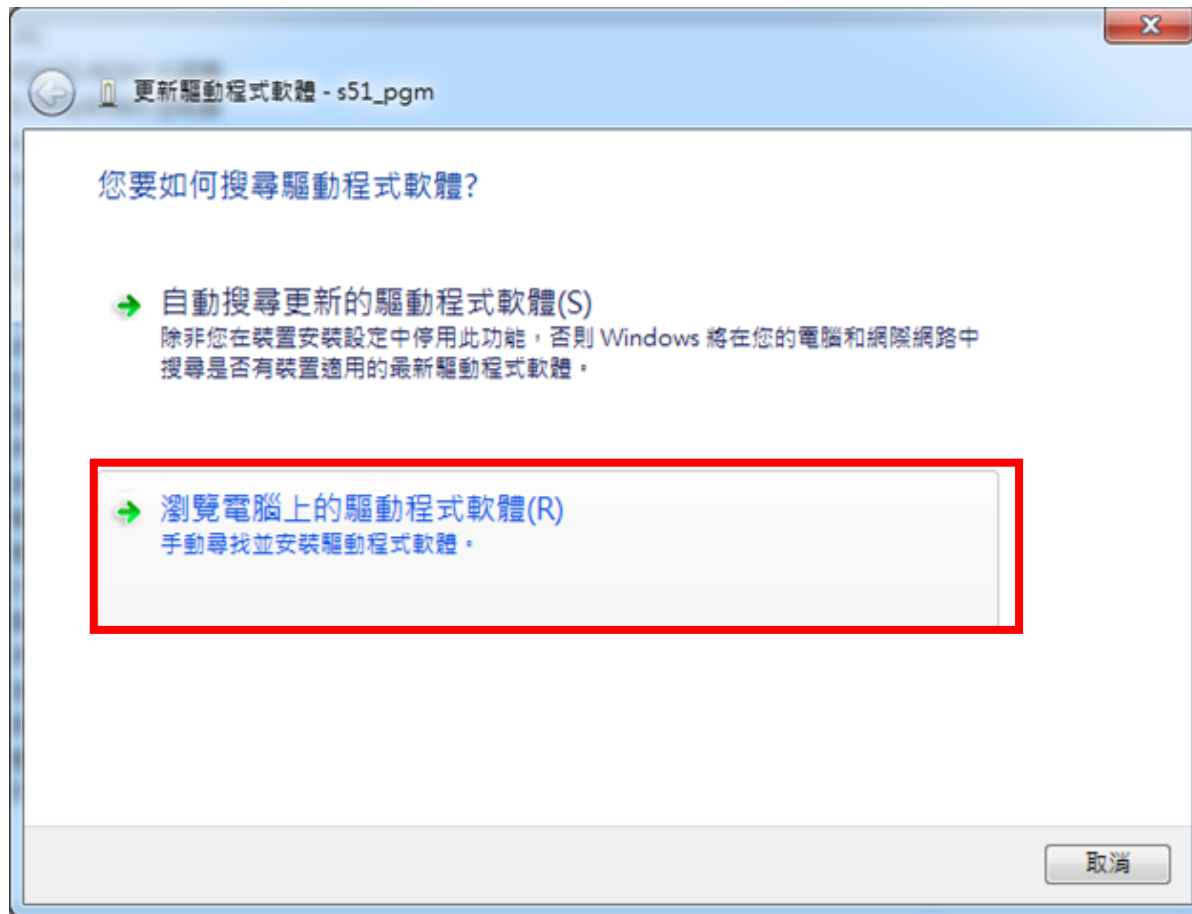
Click “裝置管理員”

USB driver installation (Win7)



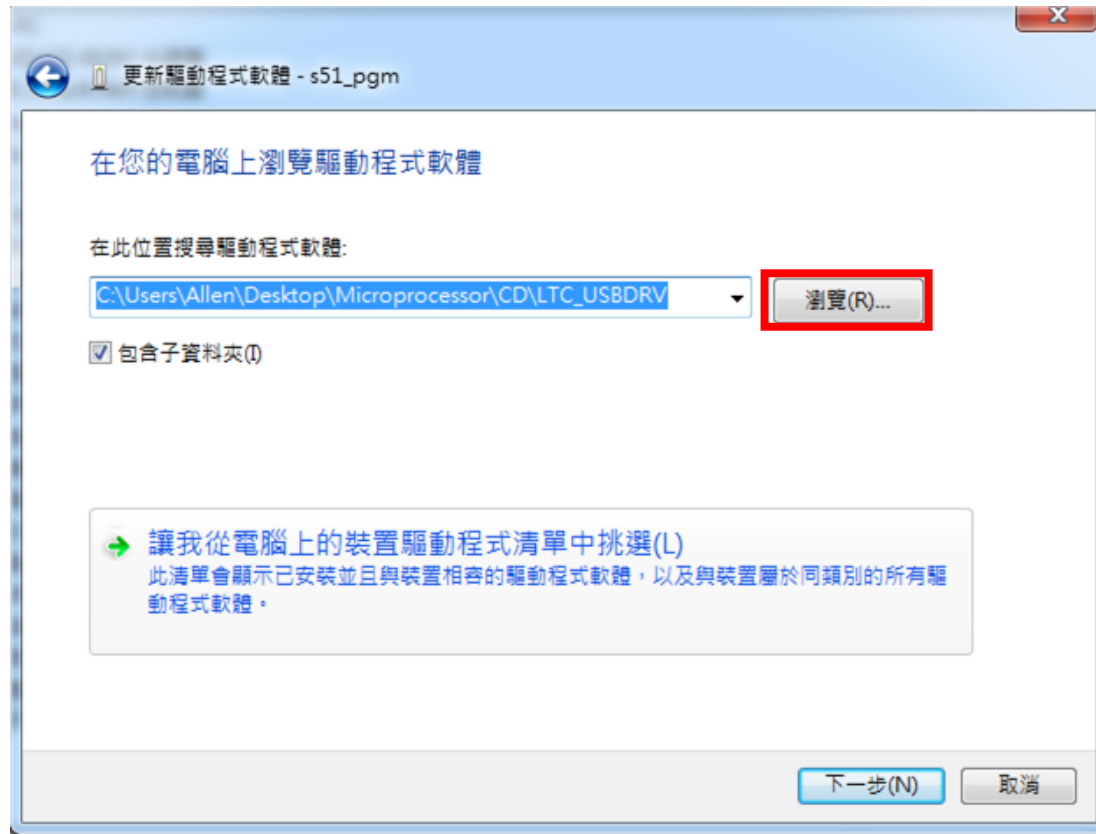
Click “s51_pgm” and right click, and then “更新驅動程式” 27

USB driver installation (Win7)



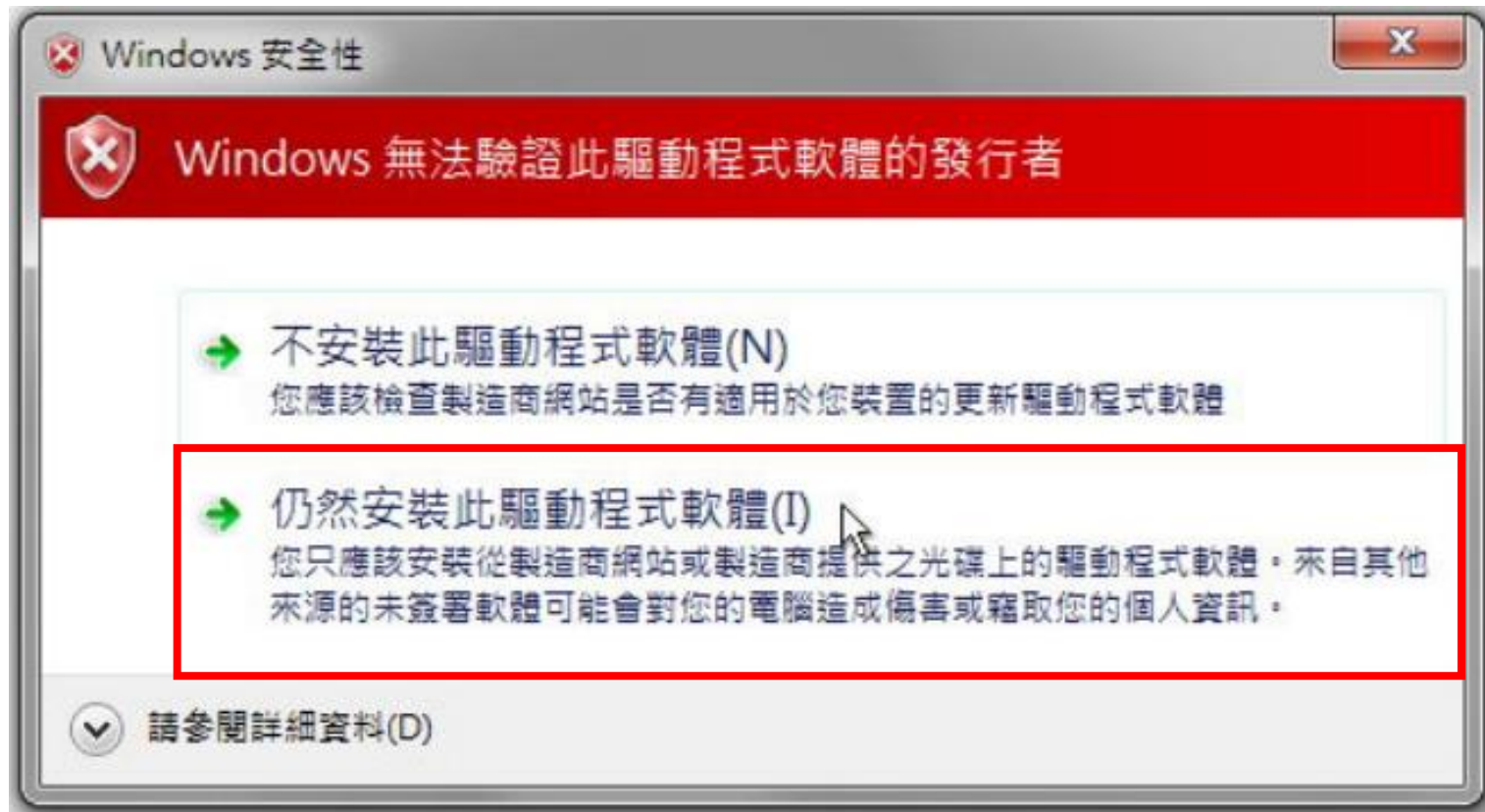
Choose “瀏覽電腦上的驅動程式軟體”

USB driver installation (Win7)



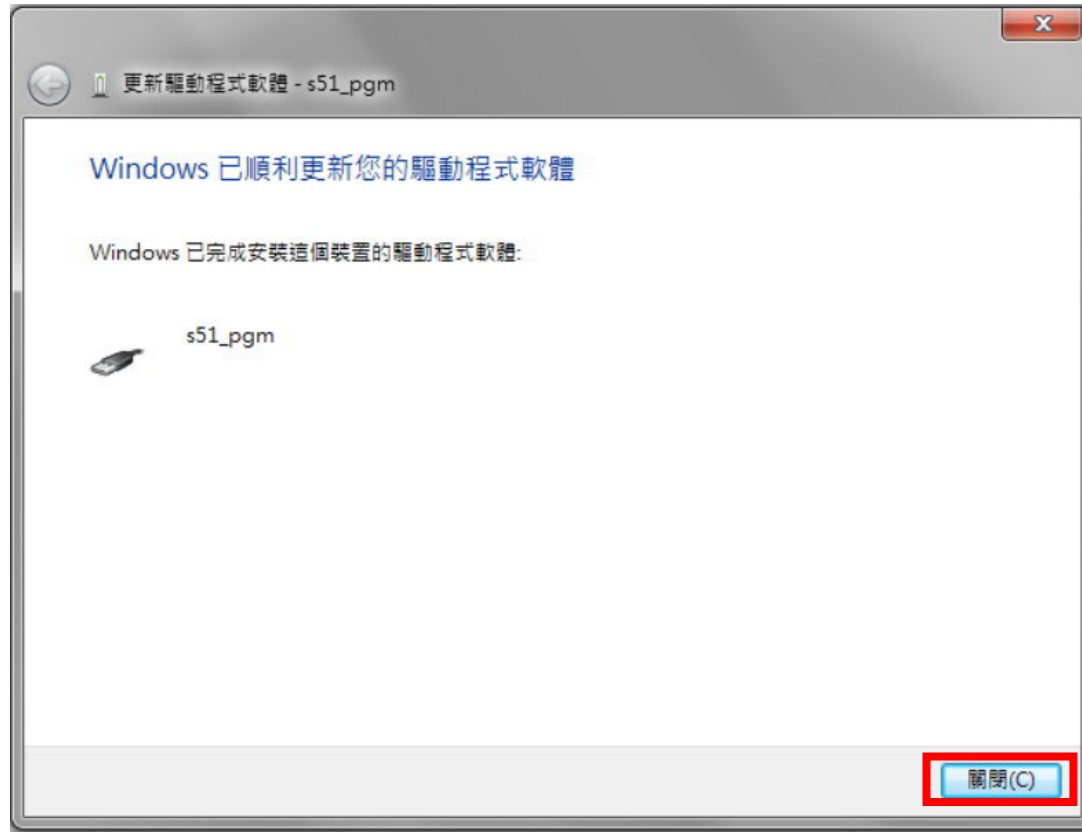
Click“瀏覽”, choose the path of USB driver
(Go to eeclass and download USB_Driver)

USB driver installation (Win7)



Choose this

USB driver installation (Win7)

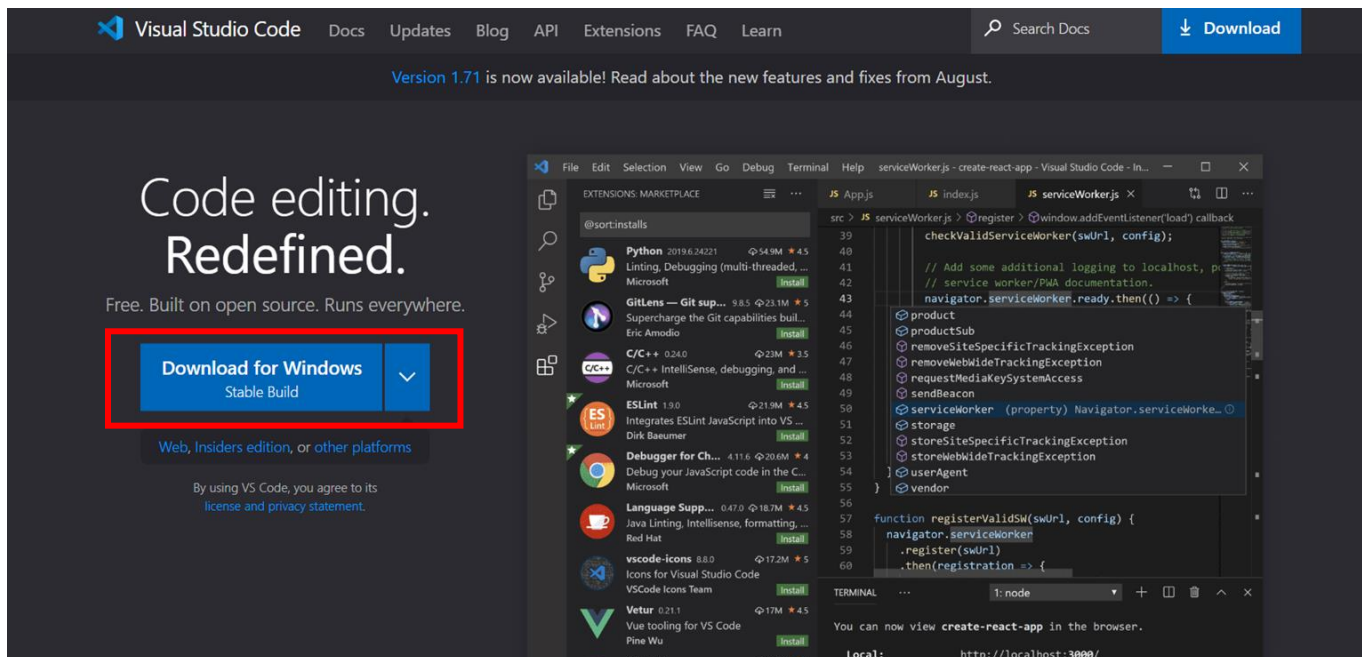


按關閉鈕關閉對話盒即可，而裝置管理員視窗裡，也將出現 s51_pgm 的驅動程式，接著重新開機

Visual Studio Code Installation

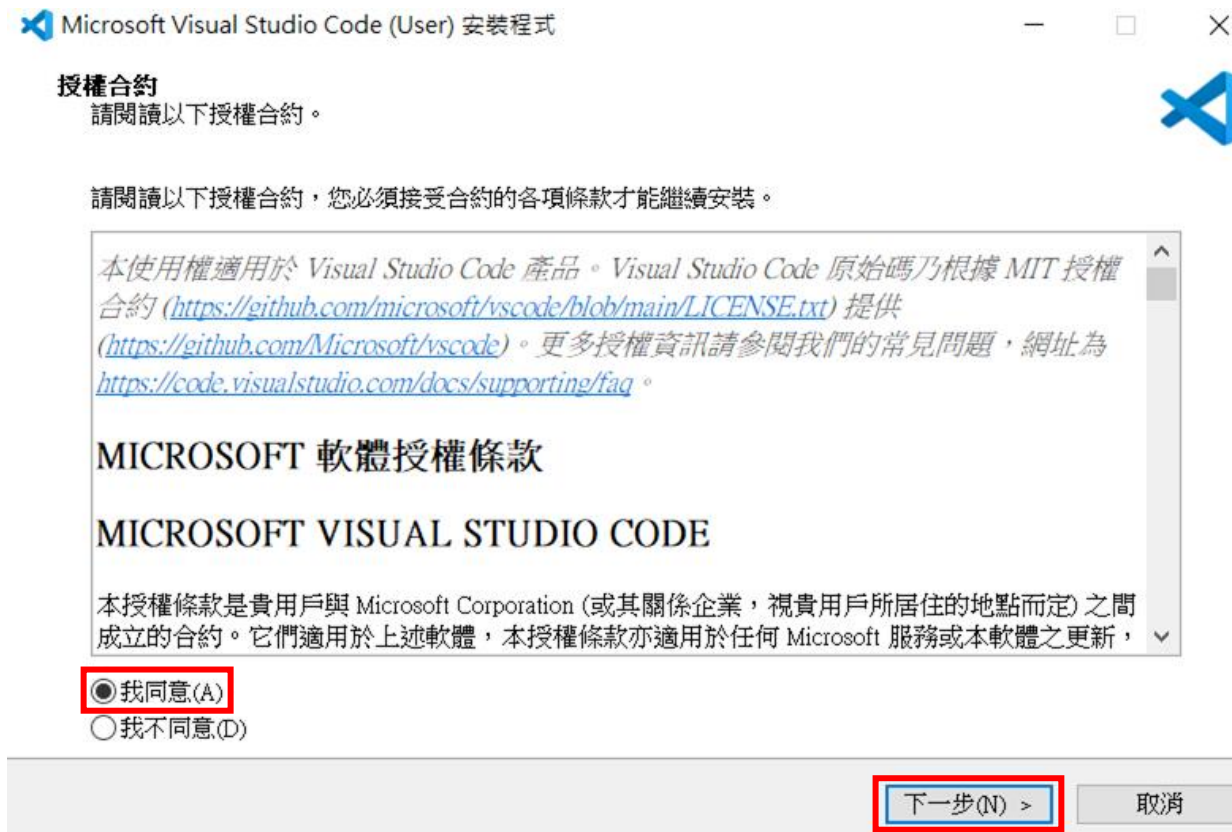
Download

- Go to VSCode Official Website
- <https://code.visualstudio.com/>
- Download it



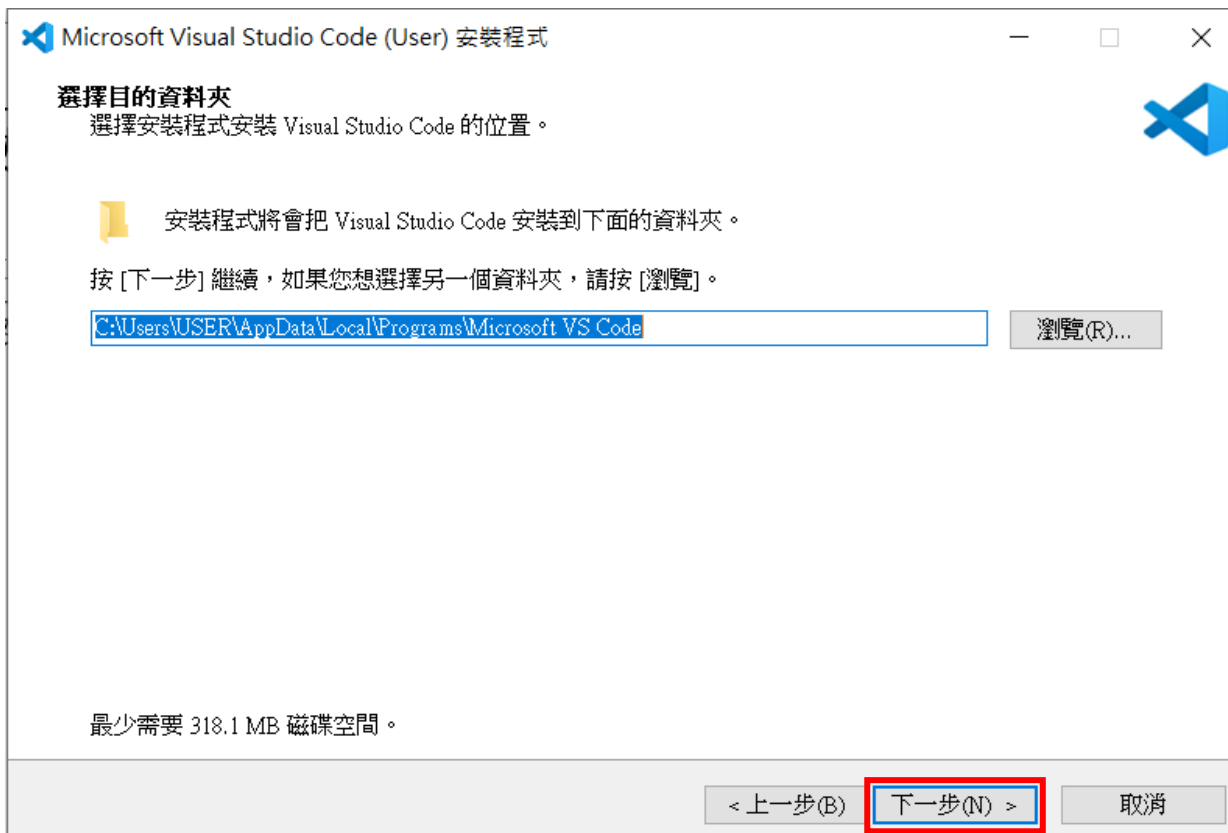
Install

- Open file

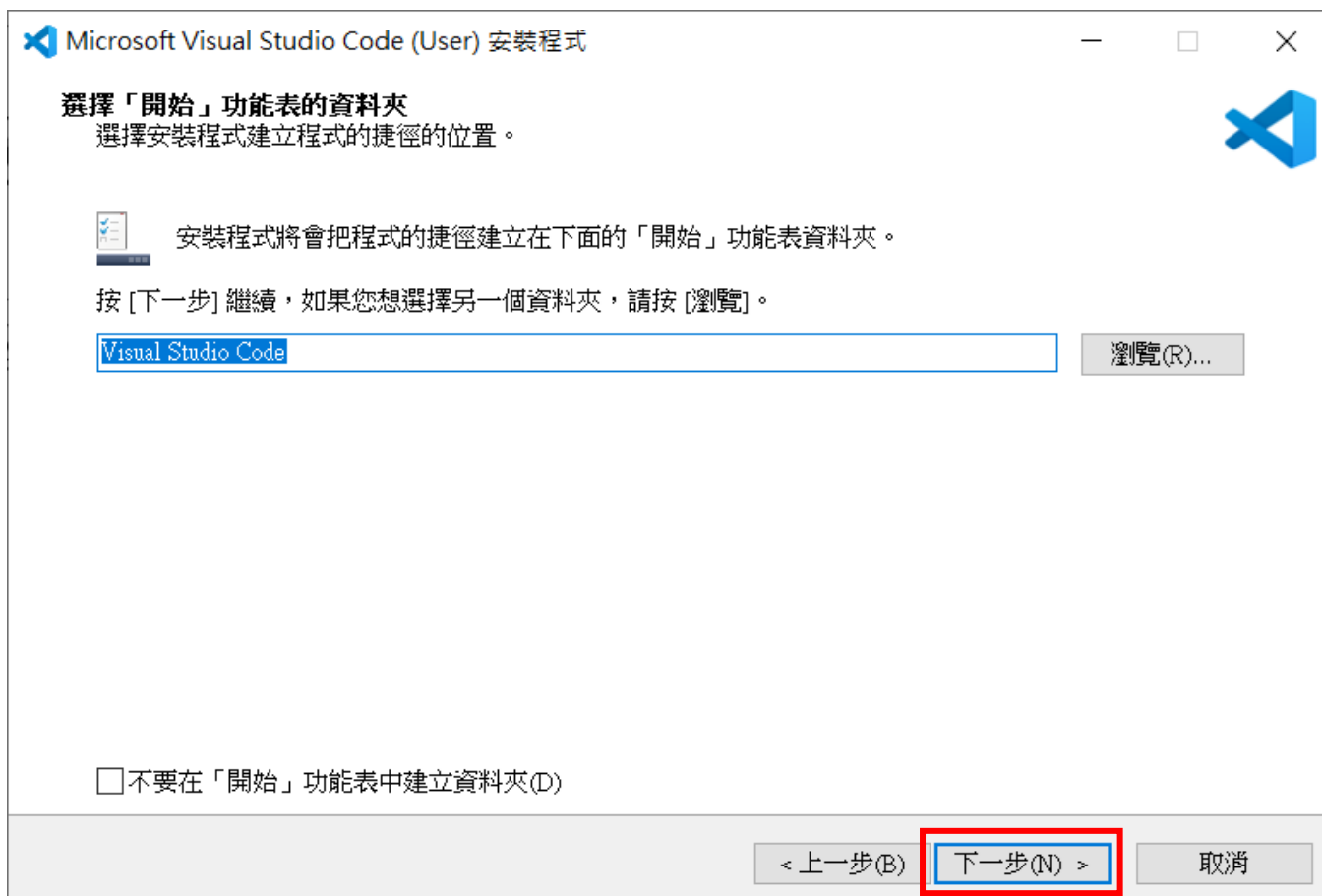


Install

- Choose the path you want. (up to you)

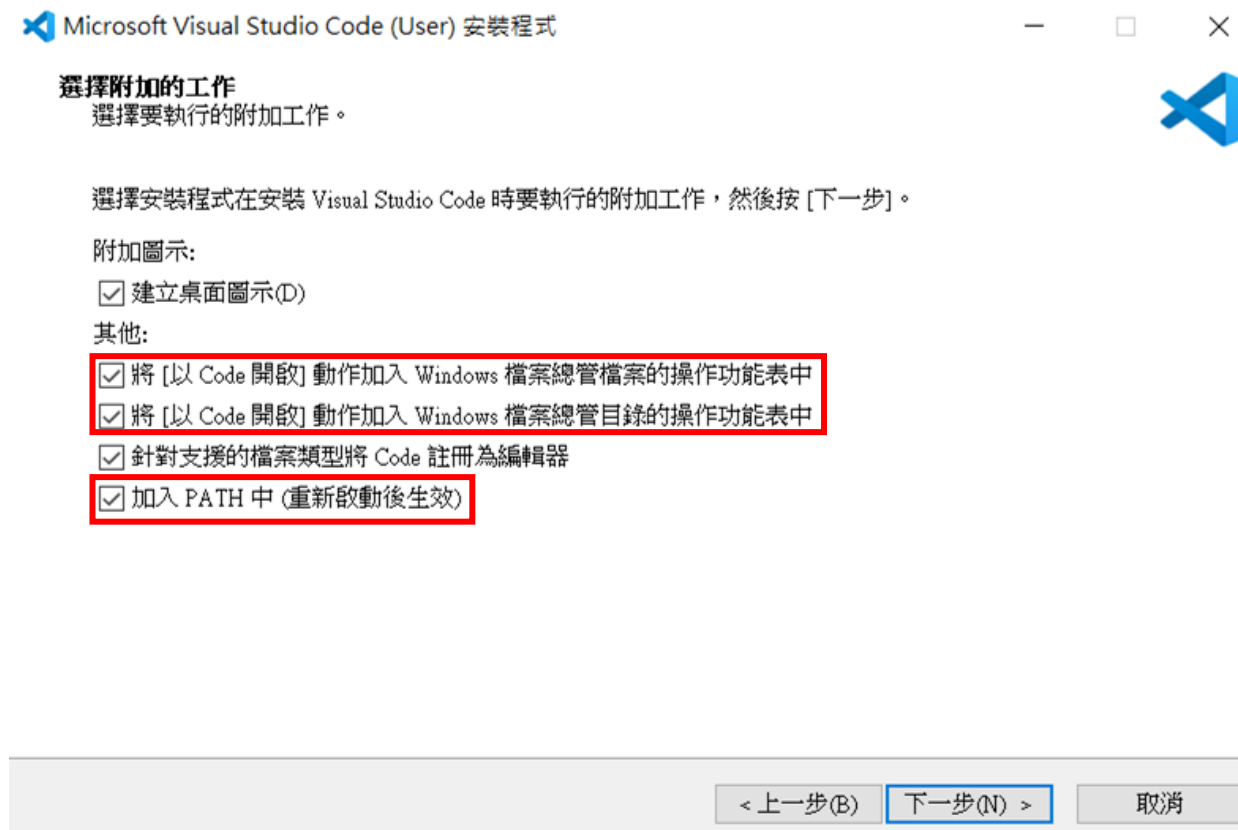


Install

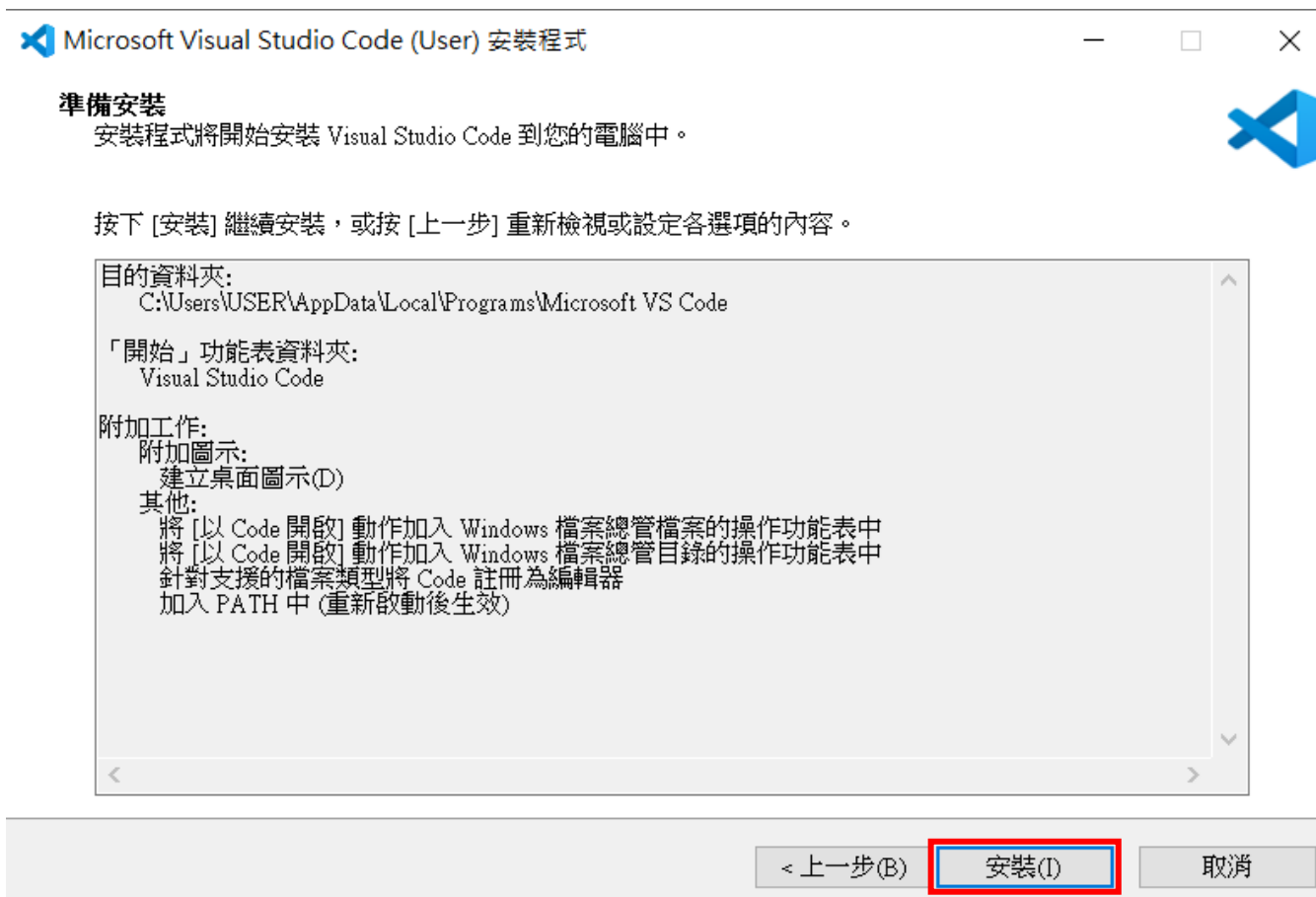


Install

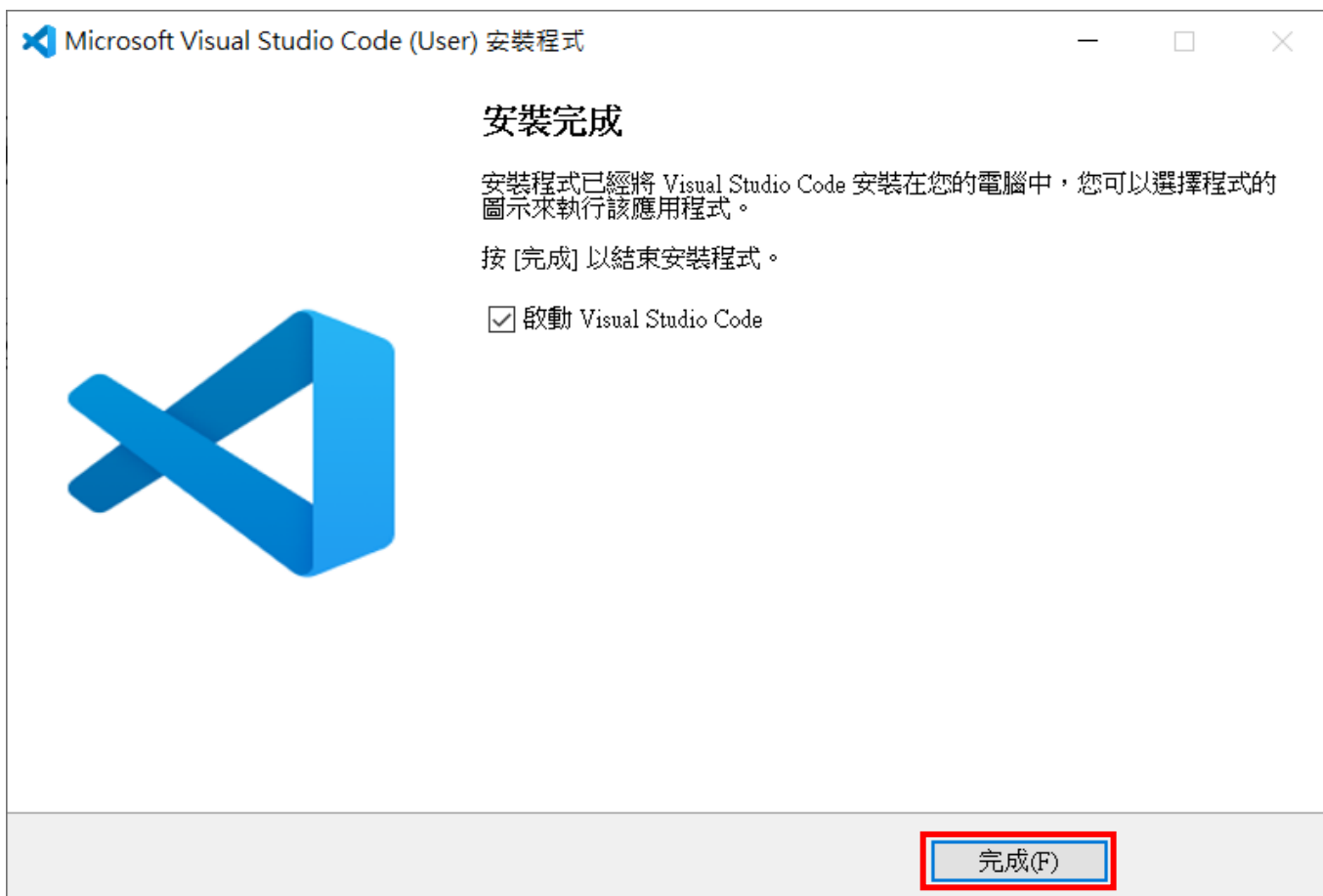
- Don't cancel highlighted checkbox !



Install



Install



SDCC Installation

Download

- Go to SDCC official website.
- <http://sdcc.sourceforge.net/>

SDCC - Small Device C Compiler

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[Source](#)

[Regression Tests](#)

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[Differences](#)

[Documentation](#)

[SDCC Manual \(pdf\)](#)

[SDCC at](#)

What is SDCC?

SDCC is a *retargettable, optimizing Standard C (ANSI C89, ISO C99, ISO C11) compiler suite* that targets the **Intel MCS51** based microprocessors (**8031, 8032, 8051, 8052, etc.**), **Maxim** (formerly **Dallas**) **DS80C390** variants, **Freescale** (formerly **Motorola**) **HC08** based (**hc08, s08**), **Zilog Z80** based MCUs (**Z80, Z180, SM83, Rabbit 2000, 2000A, 3000A, TLCS-90**), **Padauk** (**pdk14, pdk15**) and **STMicroelectronics STM8**. Work is in progress on supporting the **Padauk pdk13** and **MOS 6502** targets; **Microchip PIC16** and **PIC18** targets are unmaintained. SDCC can be retargeted for other microprocessors.

SDCC suite is a collection of several components derived from different sources with different FOSS licenses. SDCC compiler suite include:

- **sdas** and **sdld**, a *retargettable assembler and linker*, based on **ASXXXX**, written by Alan Baldwin; (GPL).
- **sdcpp** *preprocessor*, based on **GCC cpp**; (GPL).
- **ucsim** *simulators*, originally written by Daniel Drotos; (GPL).
- **sdcdb** *source level debugger*, originally written by Sandeep Dutta; (GPL).
- **sdbinutils** *library archive utilities*, including **sdar**, **sdranlib** and **sdnm**, derived from GNU Binutils; (GPL)
- **SDCC run-time libraries**; (GPL+LE). Pic device libraries and header files are derived from Microchip header (.inc) and linker script (.lkr) files. Microchip requires that "The header files should state that they are only to be used with authentic Microchip devices" which makes them incompatible with the GPL.
- **gcc-test** *regression tests*, derived from **gcc-testsuite**; (no license explicitly specified, but since it is a part of GCC is probably GPL licensed)
- **packihx**; (public domain)
- **makebin**; (zlib/libpng License)
- **sdcc C compiler**, originally written by Sandeep Dutta; (GPL). Some of the features include:

Download

Supported Windows - x86_64 Binaries



Windows 64bit package with installer (x86_64-w64-mingw32-setup)

Filename	Size	Snapshot Date
sdcc-20230917-14350-x64-setup.exe	7514 K	Sep 18 07:00
sdcc-20230916-14350-x64-setup.exe	7509 K	Sep 17 06:45
sdcc-20230915-14350-x64-setup.exe	7508 K	Sep 16 18:00
sdcc-20230912-14346-x64-setup.exe	7514 K	Sep 15 10:45
sdcc-20230911-14344-x64-setup.exe	7514 K	Sep 12 06:00



Windows 64bit (x86_64-w64-mingw32)

Filename	Size	Snapshot Date
sdcc-snapshot-x86_64-w64-mingw32-20230917-14350.zip	38353 K	Sep 18 07:00
sdcc-snapshot-x86_64-w64-mingw32-20230916-14350.zip	38353 K	Sep 17 06:45
sdcc-snapshot-x86_64-w64-mingw32-20230915-14350.zip	38353 K	Sep 16 18:00
sdcc-snapshot-x86_64-w64-mingw32-20230912-14346.zip	38353 K	Sep 15 10:45
sdcc-snapshot-x86_64-w64-mingw32-20230911-14344.zip	38356 K	Sep 12 06:00

Install



Install

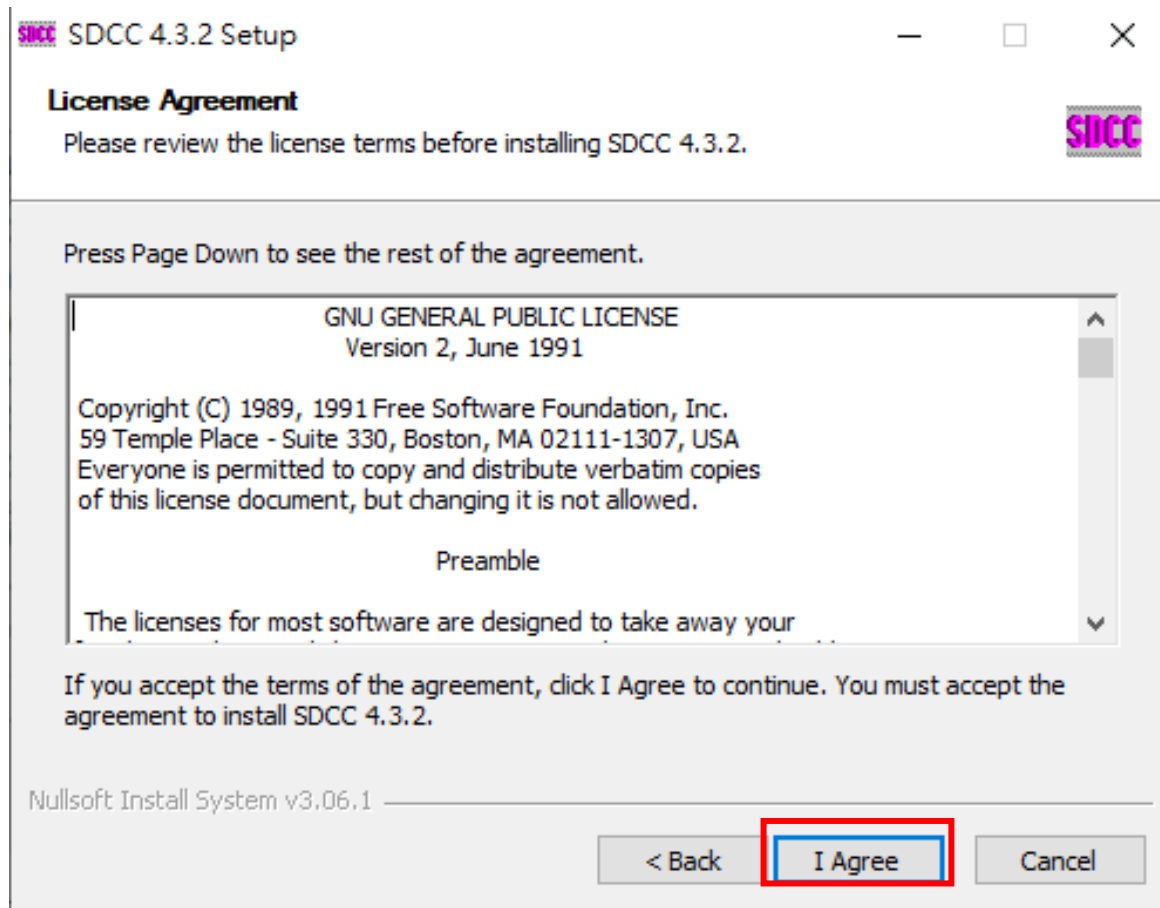


Install

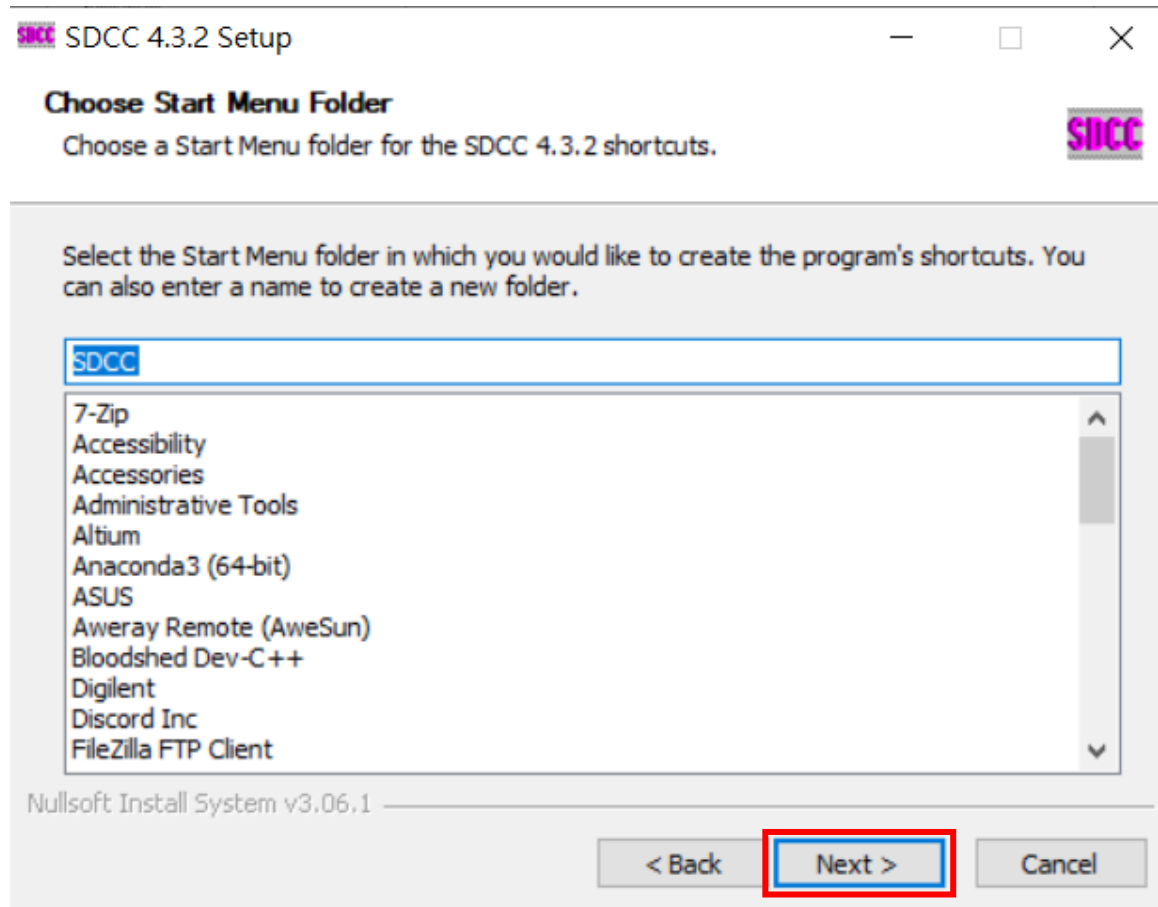
- Open the file
- Click “Next”



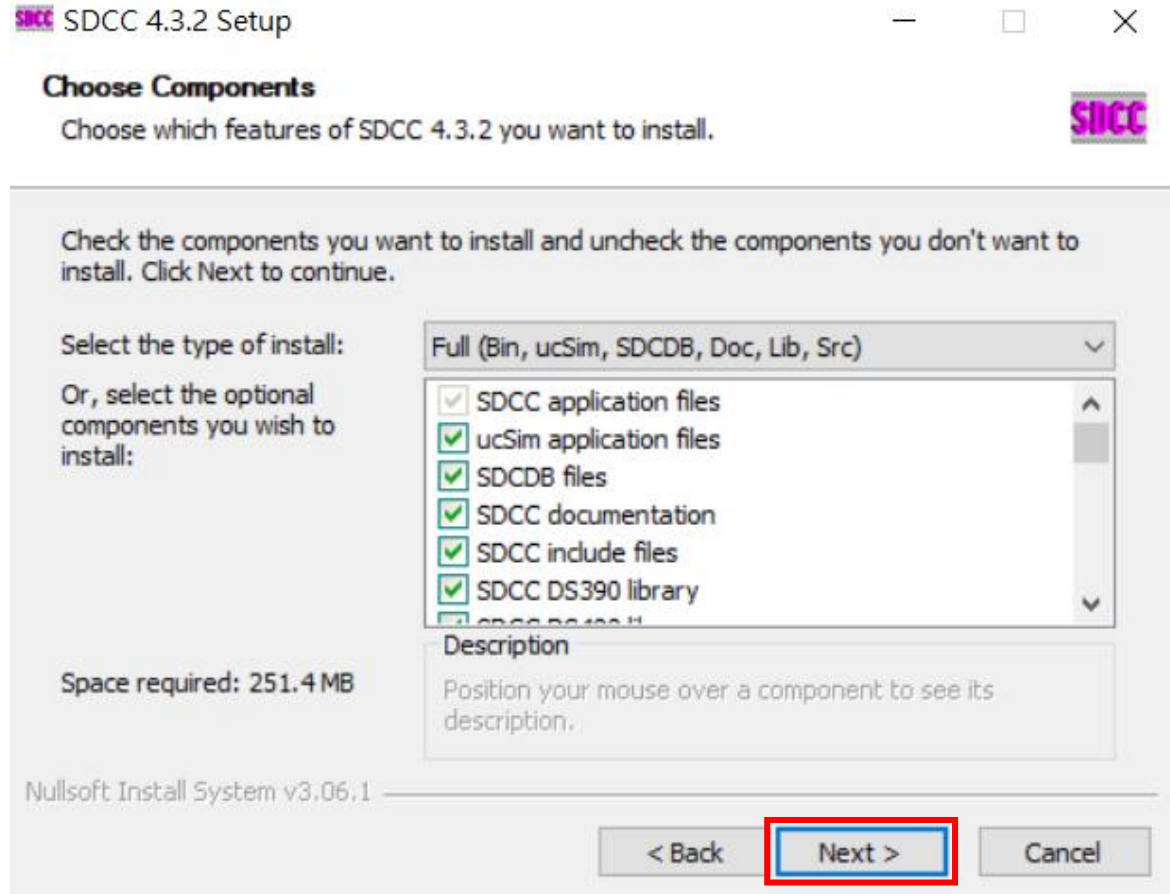
Install



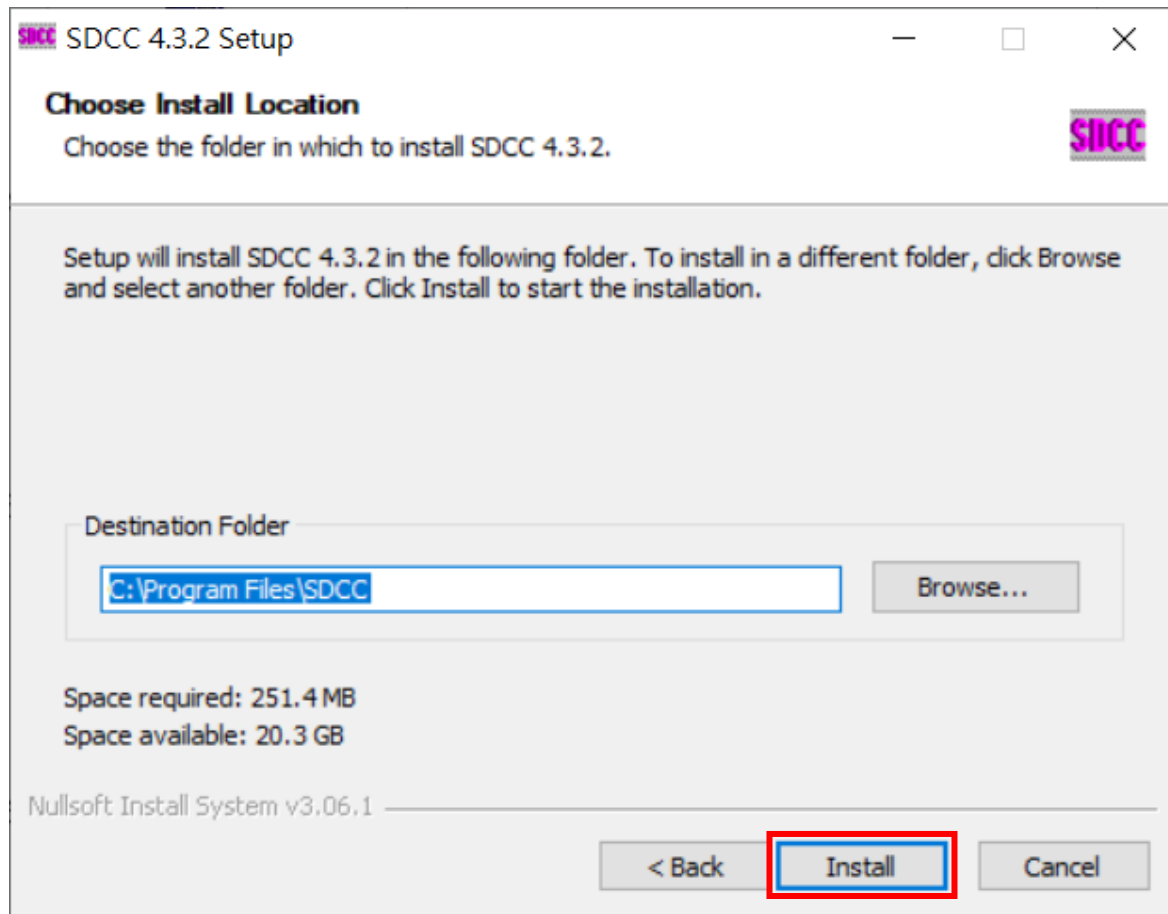
Install



Install



Install

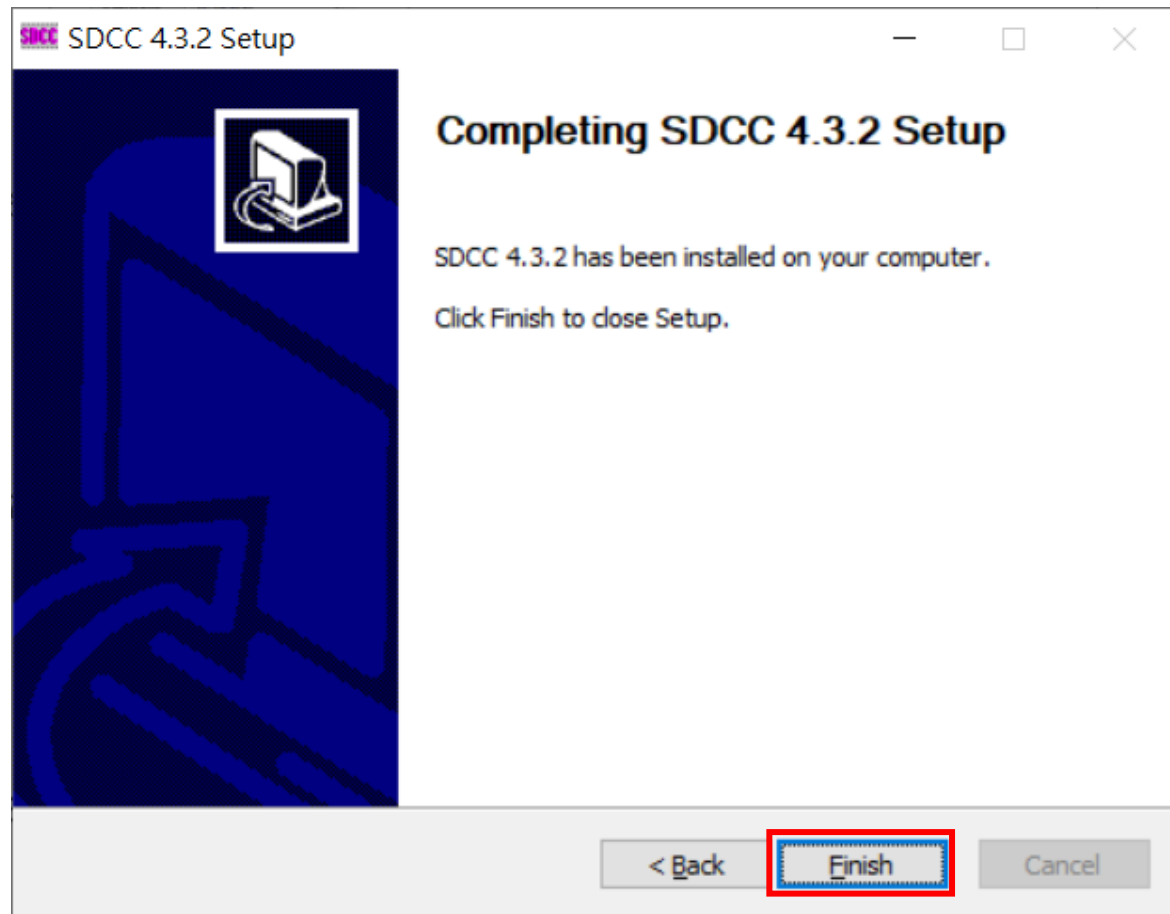


Install

- Don't cancel this checkbox



Install



MinGW Installation

Download

- Go to <https://sourceforge.net/projects/mingw/>
- Click “Download”

[Home](#) / [Browse](#) / [Development](#) / [Build Tools](#) / MinGW - Minimalist GNU for Windows



MinGW - Minimalist GNU for Windows

A native Windows port of the GNU Compiler Collection (GCC)
Brought to you by: [cstrauss](#), [earnie](#), [gressett](#), [keithmarshall](#)

★★★★★ 163 Reviews

Downloads: 5,643,385 This Week

 **Download**

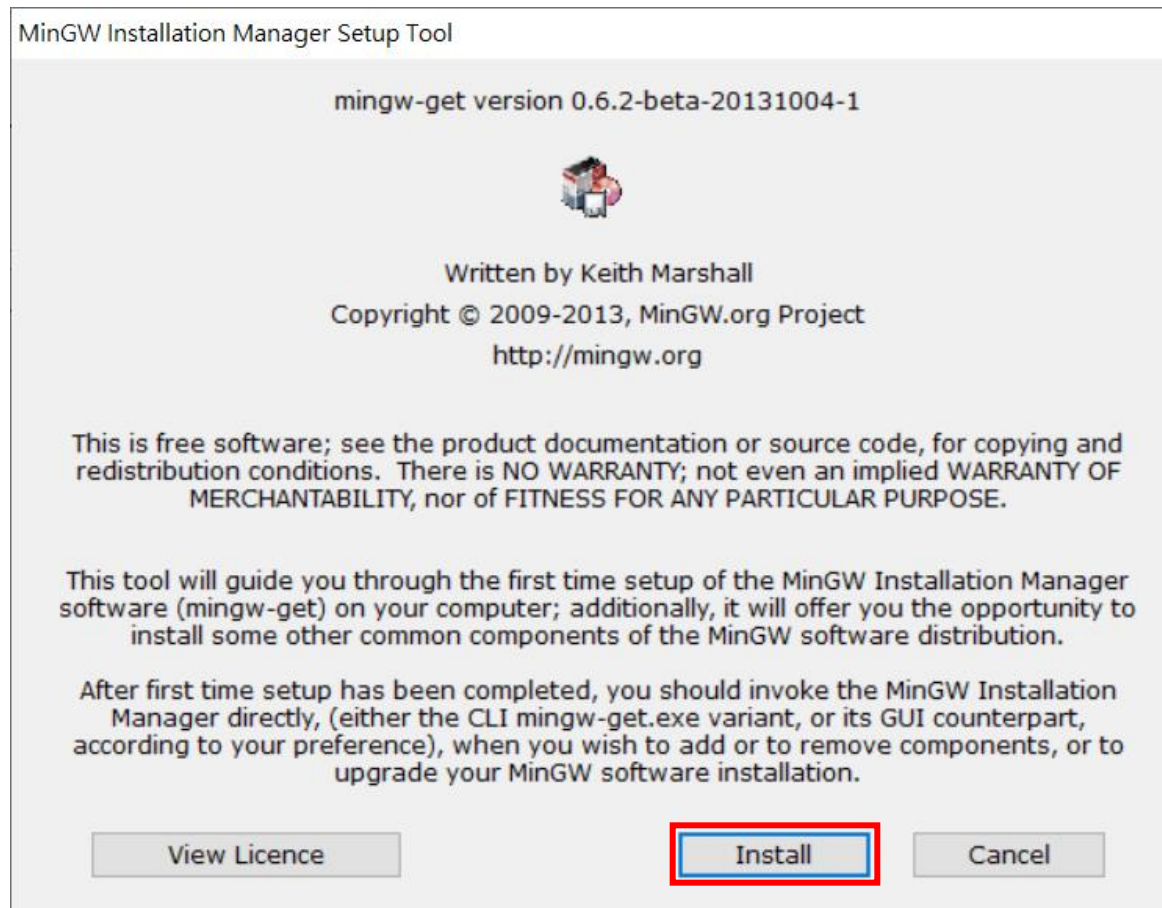


Get Updates

Share This

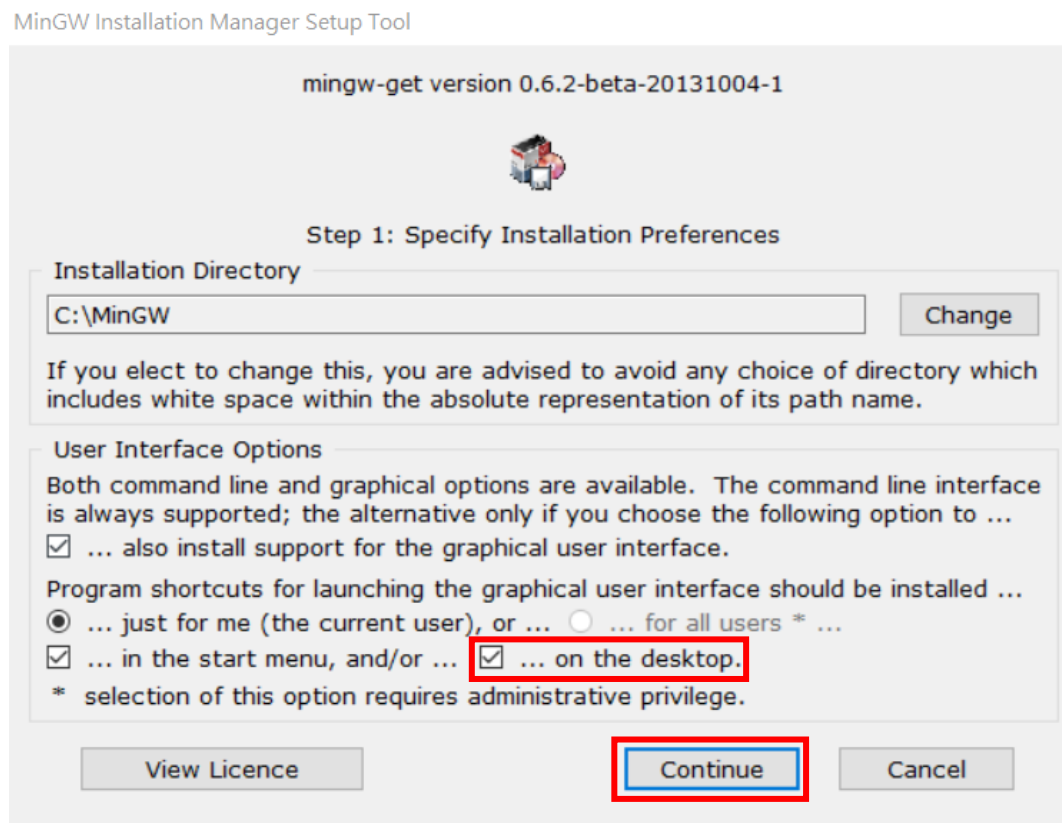
Windows

Install



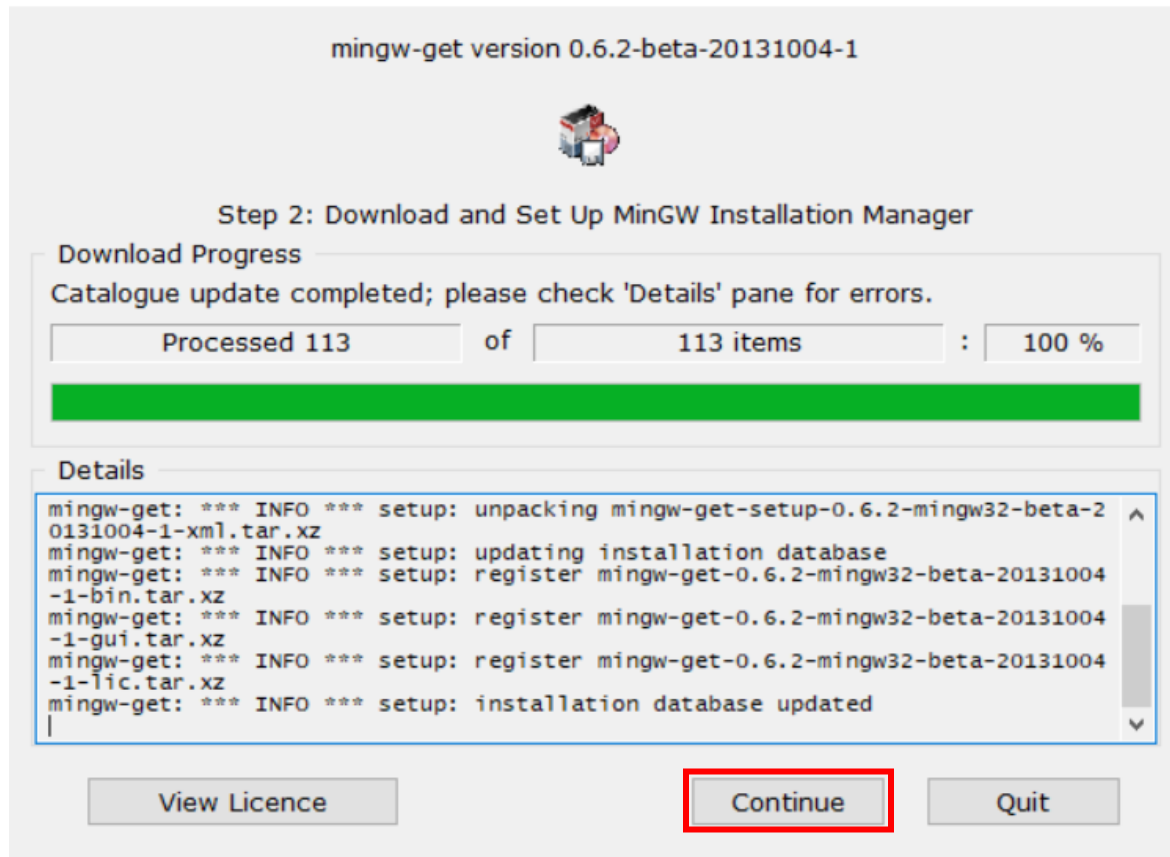
Install

- You can cancel this checkbox and then continue



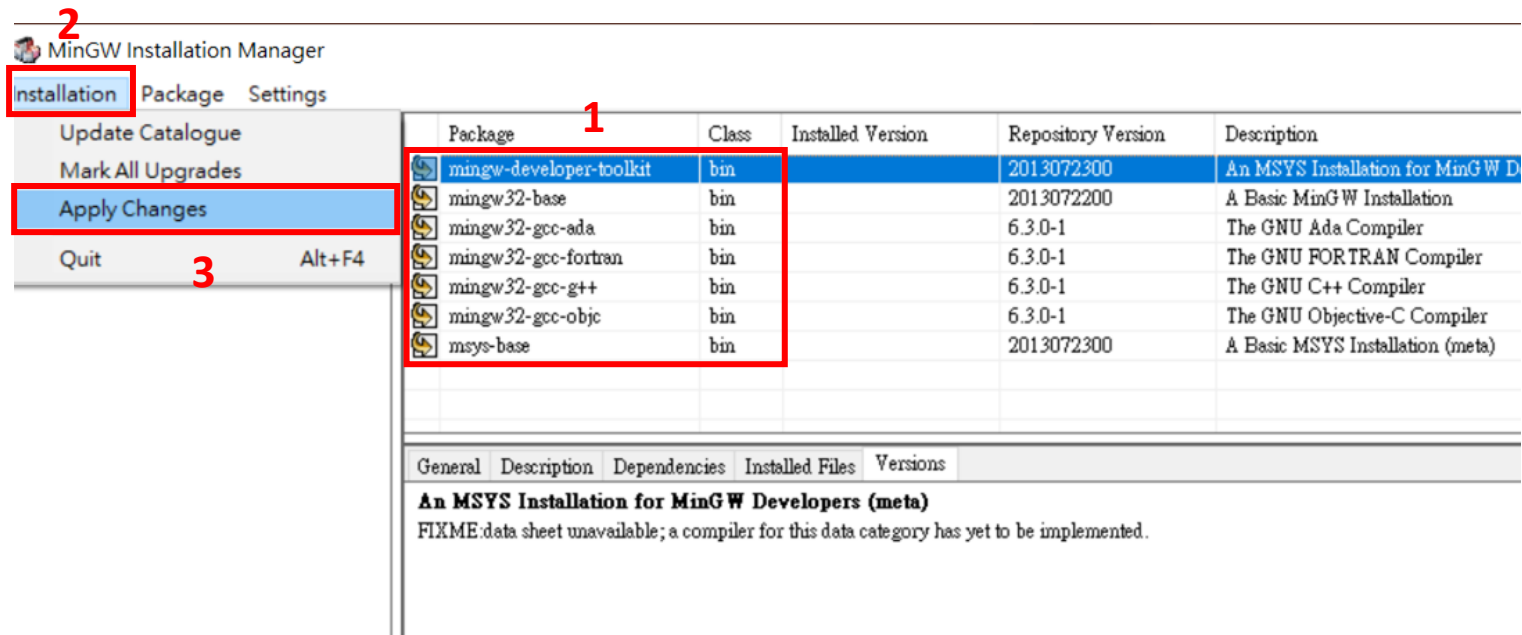
Install

MinGW Installation Manager Setup Tool



Install

- Add all checkbox
- Click “Installation”
- Click “Apply Changes”



Install

Schedule of Pending Actions

Okay to proceed?

The package changes itemised below will be implemented when you choose "Apply"

Apply Defer Discard

0 installed packages will be removed

0 installed packages will be upgraded

122 new/upgraded packages will be installed

- msysCORE-1.0.19-1-msys-1.0.19-doc.tar.xz
- termcap-0.20050421_1-2-msys-1.0.13-bin.tar.lzma
- libguile-1.8.7-2-msys-1.0.15-rtm.tar.lzma
- libregex-1.20090805-2-msys-1.0.13-dll-1.tar.lzma
- libtermcap-0.20050421_1-2-msys-1.0.13-dll-0.tar.lzma
- libpopt-1.15-2-msys-1.0.13-dll-0.tar.lzma

Install

Applying Scheduled Changes

All changes were applied successfully; you may now close this dialogue.

☐ Close dialogue automatically, when activity is complete.

Close

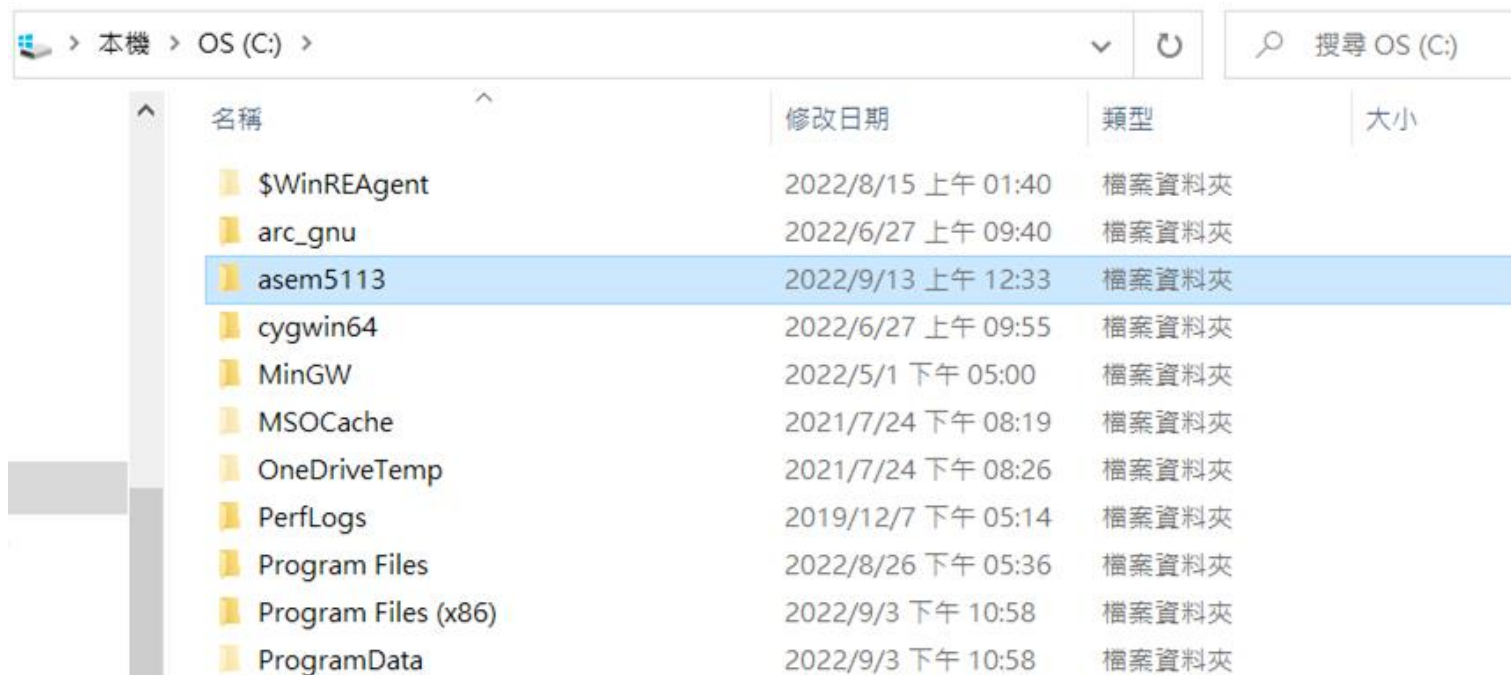
Details

```
install: libgfortran-6.3.0-1-mingw32-dll-3.tar.xz
installing libgfortran-6.3.0-1-mingw32-dll-3.tar.xz
install: gcc-fortran-4.8.2-mingw32-dev.tar.meta
installing gcc-fortran-4.8.2-mingw32-dev.tar.meta
install: gcc-objc-4.8.2-mingw32-dev.tar.meta
installing gcc-objc-4.8.2-mingw32-dev.tar.meta
install: gcc-core-6.3.0-1-mingw32-bin.tar.xz
installing gcc-core-6.3.0-1-mingw32-bin.tar.xz
install: mingw32-base-2013072200-mingw32-bin.meta
installing mingw32-base-2013072200-mingw32-bin.meta
install: gcc-ada-4.8.2-mingw32-dev.tar.meta
installing gcc-ada-4.8.2-mingw32-dev.tar.meta
install: gcc-ada-6.3.0-1-mingw32-bin.tar.xz
installing gcc-ada-6.3.0-1-mingw32-bin.tar.xz
install: gcc-fortran-6.3.0-1-mingw32-bin.tar.xz
installing gcc-fortran-6.3.0-1-mingw32-bin.tar.xz
install: gcc-c++-4.8.2-mingw32-dev.tar.meta
installing gcc-c++-4.8.2-mingw32-dev.tar.meta
install: gcc-c++-6.3.0-1-mingw32-bin.tar.xz
installing gcc-c++-6.3.0-1-mingw32-bin.tar.xz
install: libobjc-6.3.0-1-mingw32-dll-4.tar.xz
installing libobjc-6.3.0-1-mingw32-dll-4.tar.xz
install: gcc-objc-6.3.0-1-mingw32-bin.tar.xz
installing gcc-objc-6.3.0-1-mingw32-bin.tar.xz
```

ASEM-51 Installation

Unzip and Move

- Unzip “asem5113.zip” and move to C:\



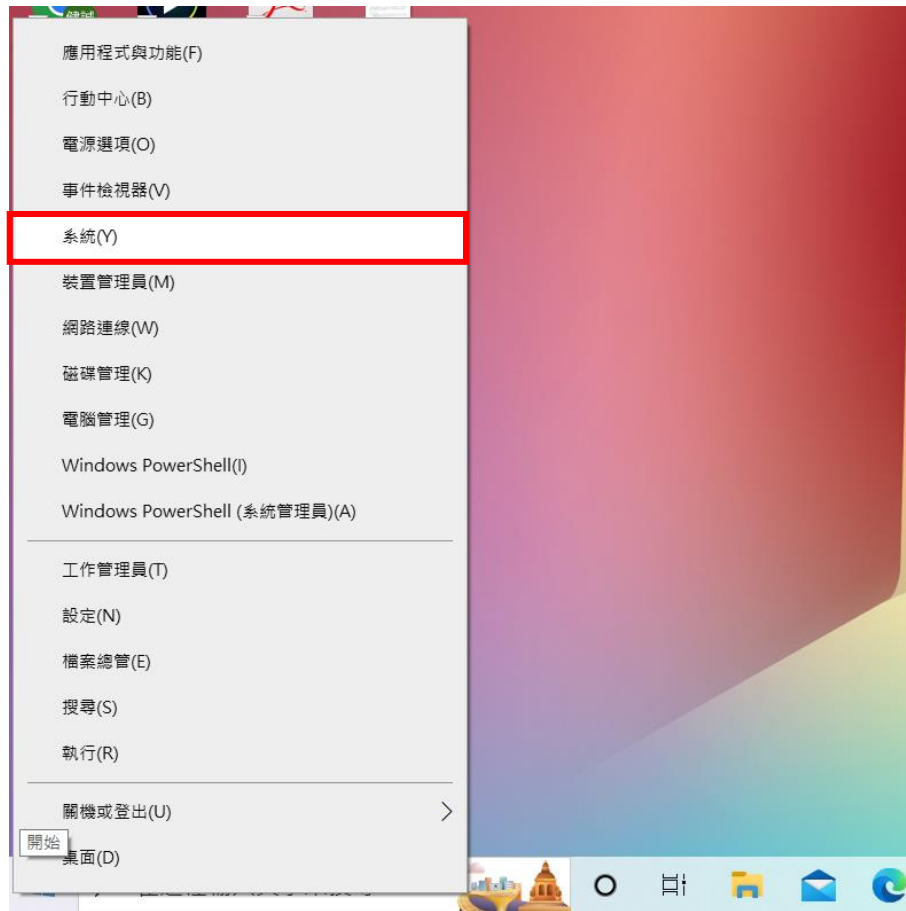
The screenshot shows a Windows File Explorer window with the address bar set to 'C:\'. The left sidebar shows the 'This PC' view. The main pane displays a list of folders in C:\, with 'asem5113' selected and highlighted in blue. The table below represents the data shown in the screenshot.

名稱	修改日期	類型	大小
\$WinREAgent	2022/8/15 上午 01:40	檔案資料夾	
arc_gnu	2022/6/27 上午 09:40	檔案資料夾	
asem5113	2022/9/13 上午 12:33	檔案資料夾	
cygwin64	2022/6/27 上午 09:55	檔案資料夾	
MinGW	2022/5/1 下午 05:00	檔案資料夾	
MSOCCache	2021/7/24 下午 08:19	檔案資料夾	
OneDriveTemp	2021/7/24 下午 08:26	檔案資料夾	
PerfLogs	2019/12/7 下午 05:14	檔案資料夾	
Program Files	2022/8/26 下午 05:36	檔案資料夾	
Program Files (x86)	2022/9/3 下午 10:58	檔案資料夾	
ProgramData	2022/9/3 下午 10:58	檔案資料夾	

Add to Path

Add to Path

- Right click “Windows” and click “系統”



Add to Path

相關設定

[BitLocker 設定](#)

[裝置管理員](#)

[遠端桌面](#)

[系統保護](#)

[進階系統設定](#)

[重新命名此電腦 \(進階\)](#)



[取得協助](#)



[提供意見反應](#)

Add to Path



Add to Path

- Choose “Path”
- Click “編輯”

環境變數

USER 的使用者變數(U)

變數	值
OneDrive	C:\Users\USER\OneDrive - 國立羅東高級中學
OneDriveCommer...	C:\Users\USER\OneDrive - 國立羅東高級中學
Path	C:\Users\USER\AppData\Local\Microsoft\Wi...
TEMP	C:\Users\USER\AppData\Local\Temp
TMP	C:\Users\USER\AppData\Local\Temp

新增(N)...

編輯(E)...

刪除(D)

系統變數(S)

變數	值
DriverData	C:\Windows\System32\Drivers\DriverData
NUMBER_OF_PRO...	8
OS	Windows_NT
Path	C:\Windows\system32;C:\Windows;C:\Wind...
PATHEXT	.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;...

新增(W)...

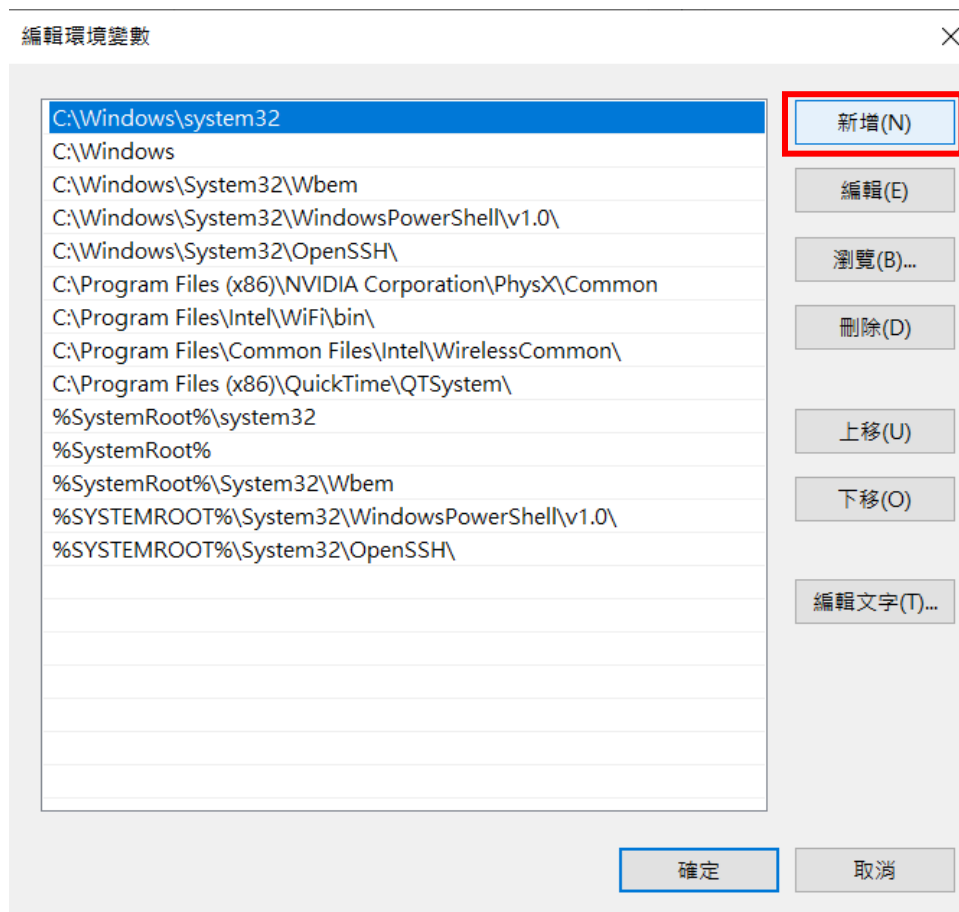
編輯(I)...

刪除(L)

確定

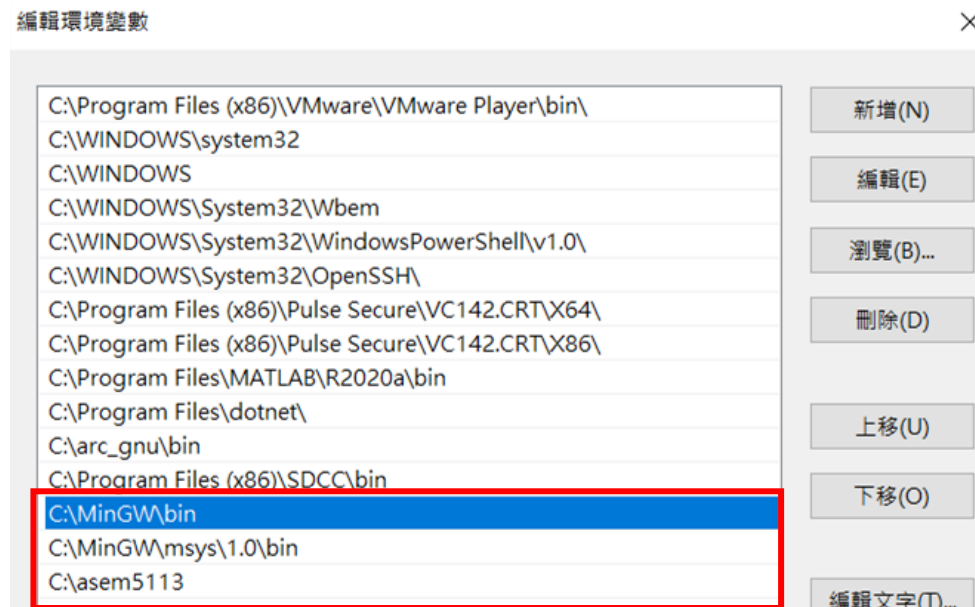
取消

Add to Path



Add to Path

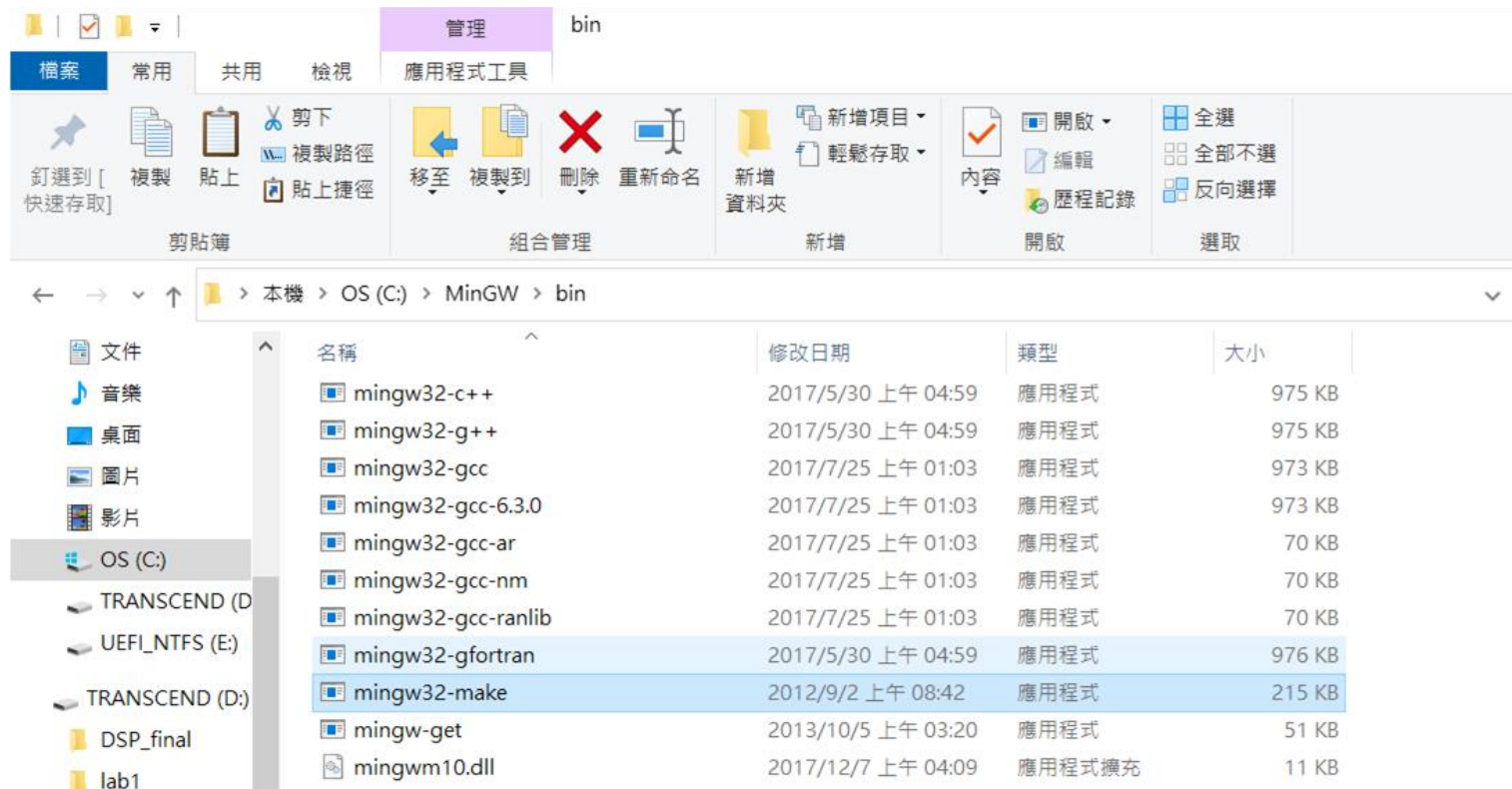
- Add “C:\MinGW\bin”
- Add “C:\MinGW\msys\1.0\bin”
- Add “C:\asem5113”
- Click “確定”



Add “make” command

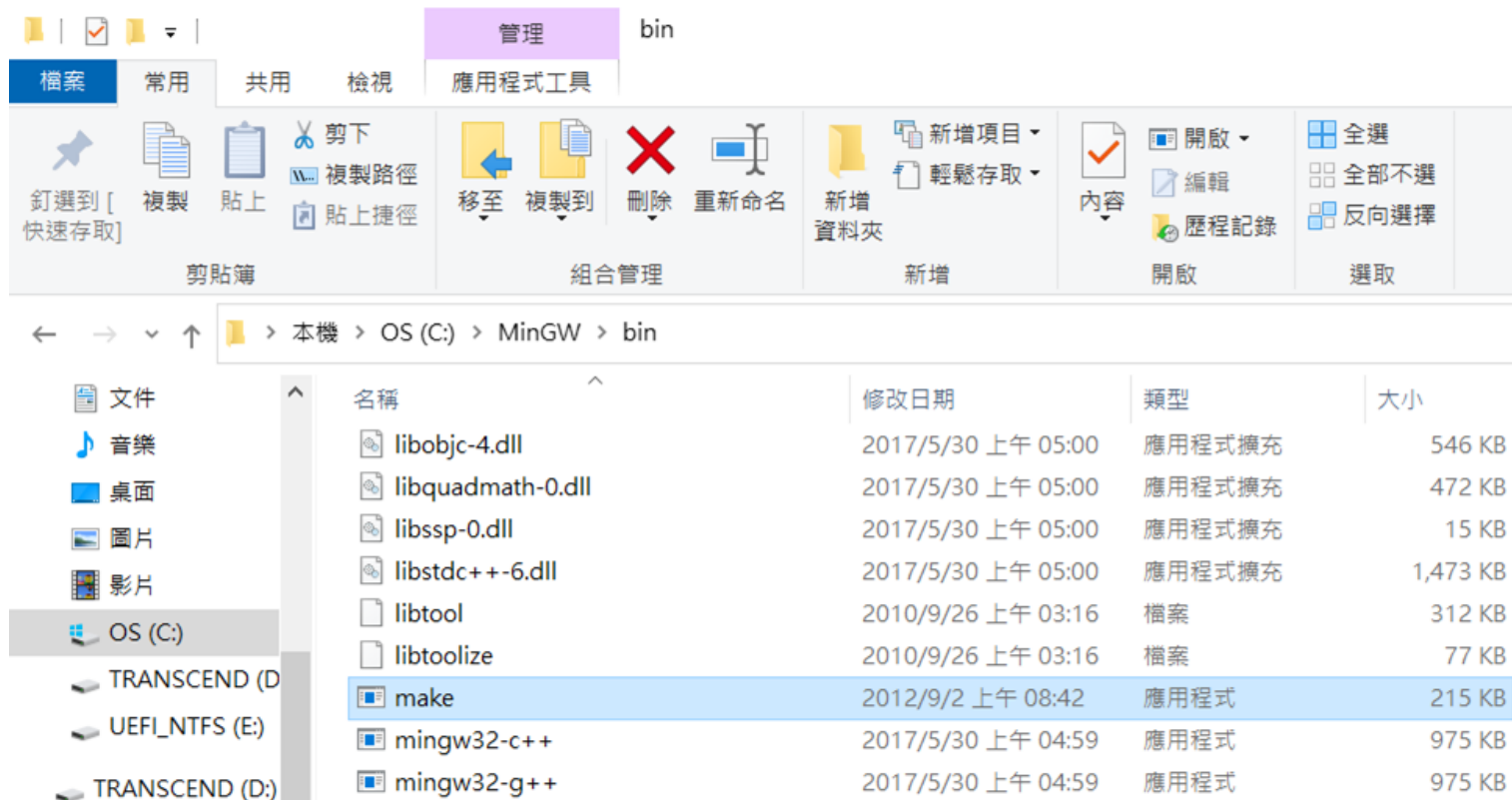
make

- Go to directory of MinGW
- Find “mingw32-make”



make

- Rename it to “make”



make

- Open cmd
- You can enter “make -v” to make sure that there exists GNU Make

 命令提示字元

```
Microsoft Windows [版本 10.0.19043.1889]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

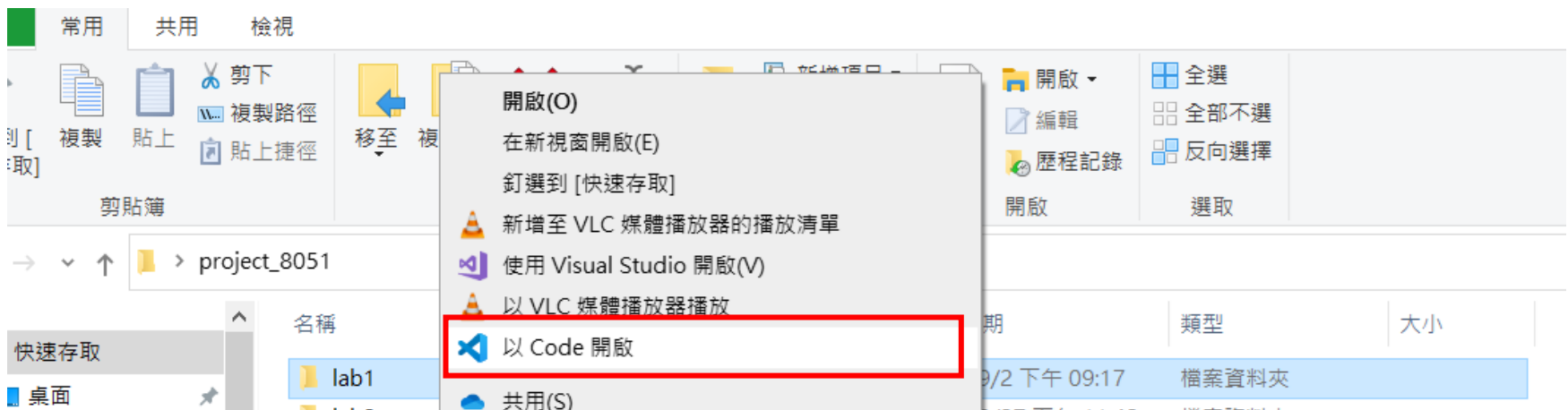
C:\Users\USER>make -v
GNU Make 3.82.90
Built for i686-pc-mingw32
Copyright (C) 1988-2012 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

C:\Users\USER>
```

How to make project

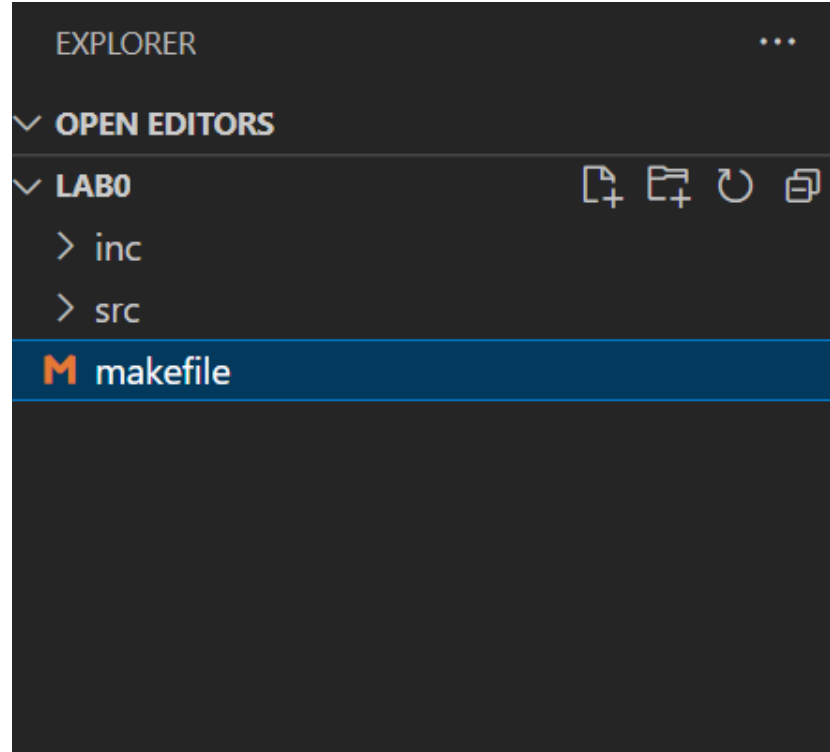
Make Project

- Open directory with VS Code



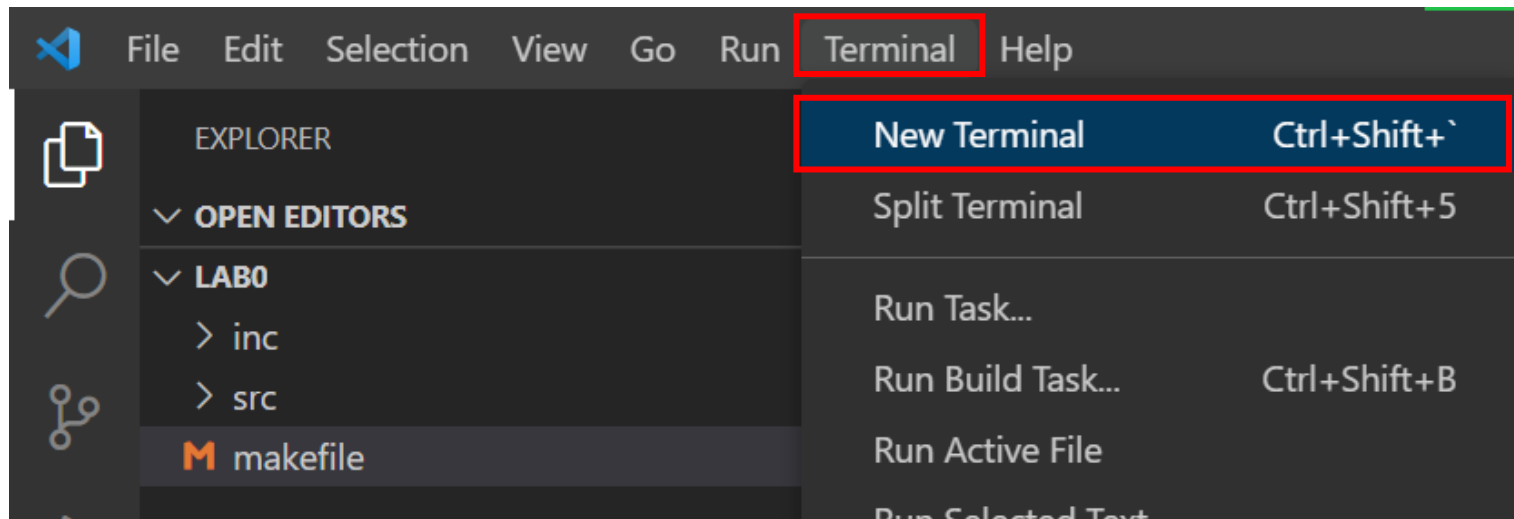
Make Project

- Look at the left-hand side
- Put “makefile” under your project directory



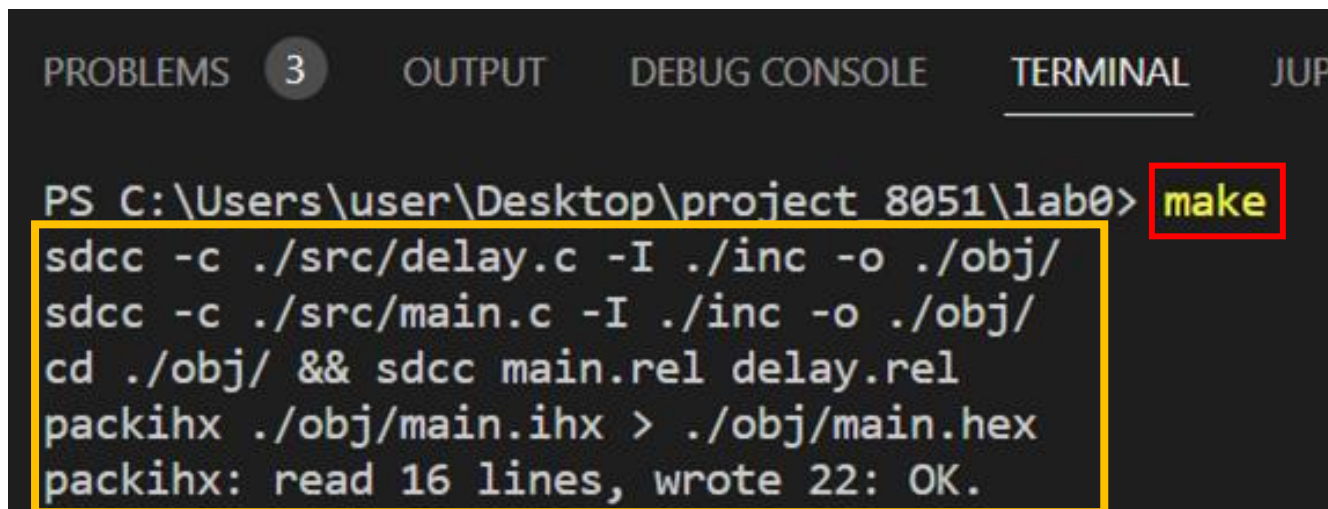
Make Project

- Click “Terminal” and then “New Terminal”.



Make Project

- Type “make” and you will see...



```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL JUP
PS C:\Users\user\Desktop\project 8051\lab0> make
sdcc -c ./src/delay.c -I ./inc -o ./obj/
sdcc -c ./src/main.c -I ./inc -o ./obj/
cd ./obj/ && sdcc main.rel delay.rel
packihx ./obj/main.ihx > ./obj/main.hex
packihx: read 16 lines, wrote 22: OK.
```



```
> inc
> obj
> src
≡ main.hex
M makefile
```

- Make sure there are not any error.
- Hex file is created!

Make Project

- Type “make clean” then remove all compile object.

```
> inc
> src
M makefile
```

```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL JUPYTER

PS C:\Users\user\Desktop\project_8051\lab0> make
sdcc -c ./src/delay.c -I ./inc -o ./obj/
sdcc -c ./src/main.c -I ./inc -o ./obj/
cd ./obj/ && sdcc main.rel delay.rel
packihx ./obj/main.ihx > ./obj/main.hex
packihx: read 16 lines, wrote 22: OK.
PS C:\Users\user\Desktop\project_8051\lab0> make clean
remove obj and main.hex
```

Be Careful!

- Only 4KB code size can be programmed into 8051

How to program .hex into 8051

Program

- After “make”, you will see “main.hex” here



The screenshot shows a file explorer window with the path 'project_8051 > lab0'. The table below lists the files and folders in the 'lab0' directory. The 'main.hex' file is highlighted with a red rectangle.

名稱	狀態	修改日期	類型
inc	✓	2022/9/13 下午 09:11	檔案資料夾
obj	✓	2022/9/14 下午 03:53	檔案資料夾
src	✓	2022/9/13 下午 09:11	檔案資料夾
main.hex	✓	2022/9/14 下午 03:53	HEX 檔案
makefile	✓	2022/9/9 上午 12:32	檔案

Program

- Open this .exe file. (in Related_File.zip)

Lab0 – Environment Setup				搜尋 Lab0 – Envir..
名稱	狀態	修改日期	類型	
Lab0-Example project	✓	2022/9/13 上午 12:19	檔案資料夾	
相關安裝軟體	✓	2022/9/13 上午 12:19	檔案資料夾	
89S51燒錄軟體	✓	2020/10/6 上午 06:28	應用程式	
Lab0 – Environment Setup	✓	2020/10/6 上午 06:28	Microsoft PowerP...	
LTC_KDM使用說明_V4.2	✓	2020/10/6 上午 06:28	Microsoft Edge P...	

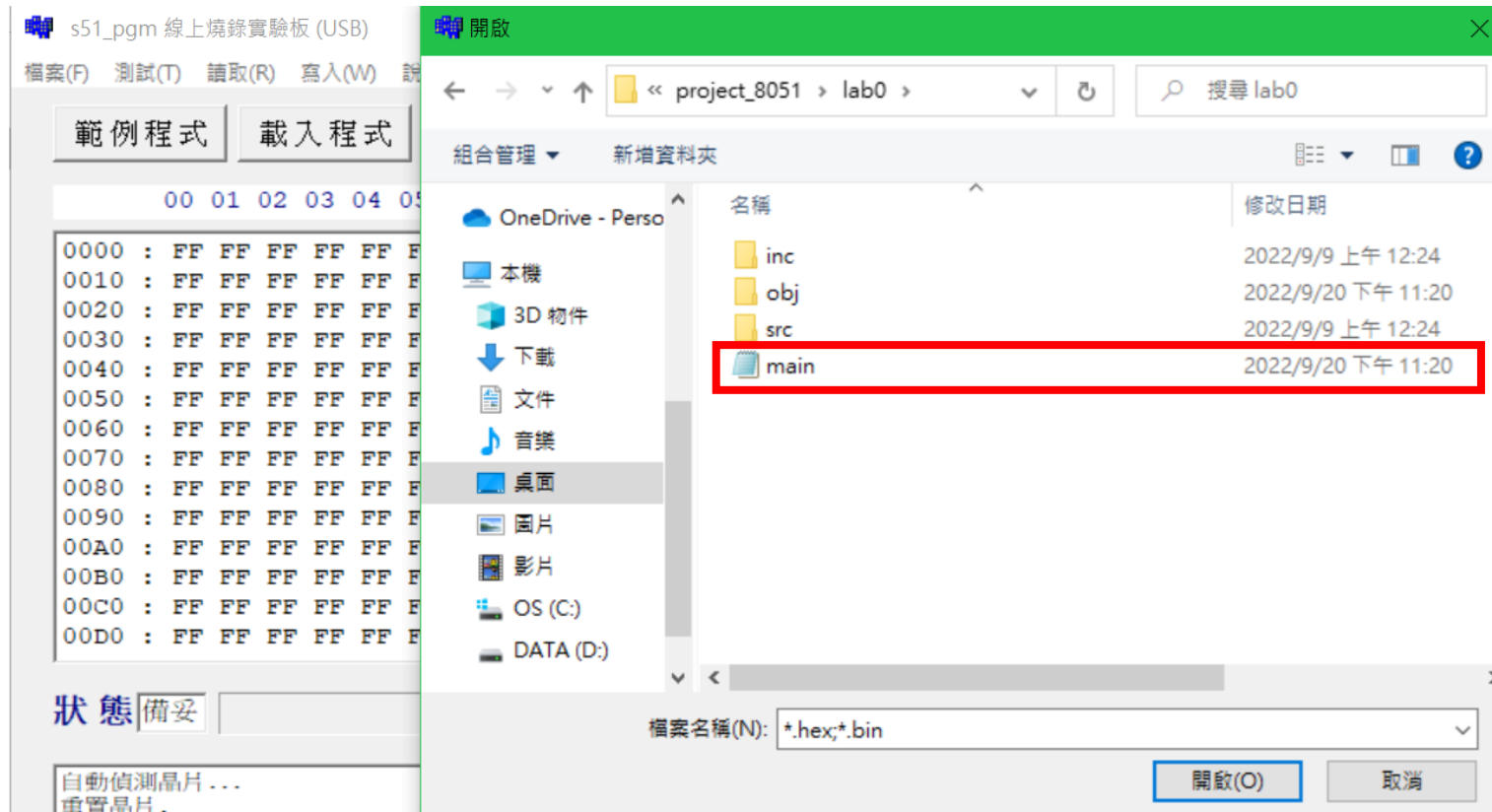
Program

- Click “載入程式”



Program

- Choose “main.hex”



Program

- Click “燒錄晶片”



Demo Video

